

CHAPTER ELEVEN

PRICE FORMATION IN FOOD RETAILING

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11.1. Introduction and learning objectives

Most agricultural commodity markets exhibit an atomistic structure and are characterised by perfect competition. Most commodities are highly homogeneous and traded all over the world. Trading is often standardised, for example on futures markets. However, food retailing, which we all experience almost daily, is very different. The food retail market is highly concentrated in many countries. The top five food retailers account for roughly 75% of the food spending by the 40 million households in Germany. In addition, food prices often display levels of dispersion over time, between products, and between stores, that exceed the costs of physical arbitrage. Finally, food products are highly differentiated. Consumers in Germany can, for example, choose between more than 2000 different yoghurt products. While many models of agricultural markets focus on single products, the food retail market clearly is a multiproduct business with manifold substitutional and complementary relationships between products. This all leads to a highly complex system of food retail pricing, many aspects of which are not fully understood. The emergence of online food retailing is adding another dimension to the system, as retailers increasingly use big data and artificial intelligence algorithms to better adjust products and prices to consumers' behaviour.

Against this background, we present some of the basic theories that have been developed to explain important characteristics of food retail pricing. In this chapter we will:

- present an overview of the structures and organisation of food retailing,

- explain the price phenomena and strategies prevalent in food retailing,
- outline the main economic models of food retail pricing, and
- summarise the main findings of several empirical case studies of food retail pricing.

In the following section (11.2), we define specific terms that are used throughout the chapter. Section 11.3 provides an overview of the food retail sector in Germany, and in section 11.4 we present various economic models and concepts that are applied to study food retail pricing. We summarise and discuss various empirical case studies in section 11.5 and close the chapter with an outlook for future research on food retail pricing in section 11.6.

11.2. Definition of terms¹

In this section, we briefly define and discuss several terms that are used in this chapter. Some of these terms will be familiar to many readers, while others might not be fully comprehensible without additional context. Hence, the reader might skip this section for the time being and return to it later for reference, if and as required.

Brand: a brand is a name, term, design, symbol or any other feature that identifies one seller's good or service as distinct from those of other sellers.

Brand loyalty: brand loyalty is “a deeply held commitment to rebuy or repatronise a preferred product/service consistently in the future, thereby causing repetitive same-brand or same brand-set purchasing, despite situational influences and marketing efforts having the potential to cause switching behaviour” (Oliver 1999, p. 34). As there is no consensus on the definition of brand loyalty, we present two additional definitions in section 11.4.

Bundling: bundling is when companies package several of their products or services together as a single combined unit, often for a lower total price than they would charge customers to buy each item separately. The products and services are usually related, but they can also consist of dissimilar items which appeal to one group of customers. **Pure bundling** occurs if only the

¹ Some of these definitions are based on <https://dictionary.cambridge.org/>.

packaged good is available; if both the packaged good and the individual goods are on sale, we refer to ***mixed bundling***.

Buying-in price: the price at which a firm can buy a good or input (purchasing or cost price).

Cartel: a cartel is a group of similar independent companies who join together to control prices, production, or other marketing activities in order to limit competition, to the advantage of the group.

Cherry picking: cherry picking refers to a consumer purchasing only products with the best price at a store, for example, only those products that are on sale at a reduced price. An extreme cherry picker is a customer who generates negative gross profits for a store on a sustained basis.

Concentration ratio (CR): the sum of the market share percentages held by the largest firms (a specified number, taken together) in an industry. The CR5, for example, measures the cumulative market share of the five largest firms in a market.

Fast-moving consumer goods (FMCG): FMCG, also known as consumer-packaged goods, are products that are sold and consumed quickly at relatively low cost with a high inventory turnover. Examples include packaged foods, beverages, toiletries, candy, cosmetics, etc.

Lerner Index: a measure of market power, the Lerner Index is the difference between the (monopoly) price and the marginal cost, divided by the (monopoly) price.

Market power: market power is the ability of a firm to profitably raise the market price of a good or service over its marginal cost. In retailing, the definition is more general: market power is the ability of a firm to influence the terms of a contract with a buyer or seller in its favour.

Mark-up: an increase in price, usually to provide a profit to the person selling. The mark-up is often measured as a percentage.

Menu costs: the costs to firms of changing prices. These costs include retraining sales staff, reprinting price lists, relabelling products and informing customers. Menu costs also include the costs of making price decisions, developing algorithms etc. High menu costs discourage firms from changing prices frequently.

Mixed strategy: in a mixed pricing strategy, a firm (randomly) sets different prices, often based on a probability distribution, in order to avoid being predictable.

Monopoly rent: the difference between the monopoly price and the marginal costs.

National brand (manufacturer brand): a brand owned and controlled by a manufacturer.

Price discrimination: price discrimination takes place when the mark-up that is charged by a firm for a product differs for different buyers, for different volumes or qualities, for different groups of buyers, or for different bundles of products, etc.

Price dispersion: variation among sellers in the price of a homogeneous good or service. Price dispersion can be interpreted as a measure of trading frictions or viewed as violation of the law of one price.

Private label (retailer brand): a brand owned and controlled by a retailer.

Promotion: activities to advertise something. Sales promotion is a set of marketing technologies aimed to stimulate the demand for a particular product and increase brand awareness.

Price promotion (sale's price): "A sale is a temporary (significant) reduction in the price of an item that is unrelated to cost changes" (Hosken and Reiffen 2001).

Reaction function: a reaction function, also called best-reply function, is the locus of optimal, i.e. profit-maximising actions that a firm may undertake in response to any given action by a rival firm. The reaction function of firm 1, for example, describes the relationship between the quantity (price) of firm 1 and the (given) quantity (price) of firm 2.

Residual demand (curve): the demand curve facing a particular firm is called the residual demand curve. The residual demand is the market demand that is not met by other firms in the industry at a given price. Hence, it is the market demand minus the supply of other firms.

Store format: retail stores can be categorised into different retail formats with respect to size, layout, product assortment, and pricing policy. The categories commonly considered in Germany are listed in Table 11.1.

Table 11.1: Food retail store formats in Germany

Store format	Size in square meters	Number of stock keeping units (SKU)	Pricing strategy*	Share of private labels
Hard discounter	<1000	1500-4000	EDLP	high
Soft discounter	<1000	2000-5000	EDLP/HiLo	medium
Small supermarket	100-399	1000-7000	HiLo	low
Large supermarket	400-999	7000-12000	HiLo	low
Small consumer market	1000-2499	12000-25000	EDLP/HiLo	medium
Large consumer market	>2500	25000-200000	EDLP/HiLo	medium

* *EDLP* = everyday low price, *HiLo* = High-Low pricing.
Source: Based on Nielsen categories (Nielsen 2018a).

Stock keeping unit (SKU): a stock keeping unit is a distinct item for sale such as a product or service and all of the attributes associated with this item that distinguish it from other items, such as package size and packaging.

11.3. Food retailing in Germany

11.3.1. History

Self-service in grocery shopping was first introduced early in the 20th century and marked a fundamental change in food retailing. In Germany, the first self-service grocery store started in 1938 in Osnabrück, and the first supermarkets opened in the 1950s. Before, the very first supermarkets had been introduced in the 1930s in the USA. They became popular around the world and are still seen as the gold standard today. Desai et al. (2013:3) call this initial period “retail 1.0”. About fifty years later, Carrefour and Walmart developed the concept of food stores which offer everything in one place and promoted the idea of one-stop-shopping, referred to as “retail 2.0”. At roughly the same time, discounters began to operate in Europe. Discount outlets typically have simple interior designs and product placement. They are further defined by narrow product assortments and **everyday low-price (EDLP) strategies**. Retailers also began to introduce **private labels** to strengthen their bargaining power vis-à-vis manufacturers, as well as to advertise their respective retail chains. At first, these brands were low-price

and low-quality, as befits the discount idea. Later, private labels began to imitate higher quality manufacturer brands and cover a larger assortment of products. What we see today is the fourth generation of private labels in food retailing.² Aided by the adoption of these innovations, retail chains quickly grew in size, which improved their bargaining power vis-à-vis food processors.

In 1990, Jeff Bezos founded Amazon and started a new era of online retailing, thus initiating the rise of e-commerce or “retail 3.0” (Desai et al. 2013:5). From an online bookstore, Amazon quickly expanded to offer music, electronic equipment and most other fast-moving consumer goods. In 2017, Amazon offered 230 million items, a number that had doubled in only three years since 2014 and that far exceeds the number of **stock keeping units (SKU)** of even the largest consumer markets (see Table 11.1).

Today we are on the brink of “retail 4.0” which is characterised by multichannel food retailing. Online retailers such as Amazon are establishing brick-and-mortar outlets, and many traditional brick-and-mortar retailers are operating online food retailing formats. Stores routinely experiment with alternative modes of distribution, such as delivery stations and home delivery. The future of retail 4.0 is expected to bring even more virtual stores, mobile payment, self-checkout, social media marketing, and dynamic pricing strategies (Desai et al. 2013).

The World Wide Web has changed consumers’ communication and information search behaviour. Over 90% of the German population has access to and uses the internet. The share of users only drops sharply (to 50%) for individuals aged 70 and above. The internet is mainly used for information searches and private e-mails (60-70%). 30% of the population uses online shopping and watches movies via streaming services (AGOF 2020). The role of information has fundamentally changed: while information was once scarce and costly to obtain, today it is abundantly available in real time, and the emphasis has shifted from finding information

² The first generation of private labels are low-price, no-name brands, as found with many discounters. Second generation private labels are quasi-brands for large packs of single products. In the third generation, private labels are used as corporate brands for groups of products, also with a view to promoting the specific retail chain. The fourth generation of private labels mimics manufacturer brands in order to compete directly with them and further improve the quality image of the retail chain. Private labels account for about 40% of the total turnover in fast moving consumer goods in Germany.

to selecting and processing relevant information (Verbeke 2005, p. 347ff). As Shapiro and Varian (1998, p. 6) put it, “nowadays the problem is not information access but information overload”.

Furthermore, information is no longer exclusively in the hands of traditional providers such as manufacturers, news agencies, and service providers. Information is a new currency that is produced, consumed, and traded at high speed (Shapiro and Varian 1998, p. 5). Consumers can also become part of the information chain by posting personal ratings on electronic marketplaces and rating platforms. As so-called influencers, some consumers participate actively in the creation and dissemination of information, using social media platforms to rate and promote goods, services and ideas (Labrecque et al. 2013).

11.3.2. Market structure and firm organisation

Food products are sold in a variety of store formats. According to Nielsen, we can distinguish between supermarkets, consumer markets, and discounters (Table 11.1). In terms of sales area, supermarkets range from 100 to 999 m² and mainly sell food products. Discounters have a narrow product assortment and mostly follow an EDLP strategy. Consumer markets are larger than supermarkets, with sales areas ranging from 1000 to more than 5000 m², and selling a broad range of food and non-food products.

Food products are also sold in drugstores, gas stations, off-licence stores, kiosks, cash-and-carry markets, specialised food stores (bakeries, butchers, fish shops, etc.) and weekly markets. In the following discussion we focus on supermarkets, consumer markets, and discounters, although drugstores are included in some statistics. In Germany, the number of stores with a sales area greater than 100 m² declined from 37,109 in 2014 to 34,947 in 2018. Their turnover increased from € 167 billion in 2013 to € 188 billion in 2017 (Nielsen 2018b). Table 11.2 shows turnover and market share figures for the different store formats in 2017. Consumer markets and discounters account for 84% of the total turnover of food stores larger than 100 m².

Table 11.2: Turnover in German food retailing in 2017

Store format	Turnover in million Euro	Share of total turnover in %
Large consumer market	45213	24.1
Small consumer market	40665	21.7
Discounter	70080	37.4
Large supermarket	13622	7.3
Small supermarket	1815	1.0
Drugstore	16108	8.6
Total	187503	100

Source: Nielsen 2018a, p. 111.

As mentioned above, the five biggest food retail chains in Germany dominate the market with a combined market share of about 74% for food products. Table 11.3 shows that if non-food sale by the food retailers is also considered, the picture does not change substantially. Edeka is the leading food retail chain in Germany with a market share of 25.7% and 376,000 employees. Netto, another chain of discounters and supermarkets, belongs to Edeka. The first Edeka cooperative was founded in 1898 in Berlin. Edeka still is a cooperative of about 4000 mostly independent merchants who run the individual stores. These stores are supplied by seven regional trading companies that trade food products and produce some of their own. The regional trading companies also consult the Edeka food stores and organise marketing, advertising, and provide financial services through a bank.

The Rewe Group started in 1927 as a cooperative and transformed into a non-listed stock company in 1972. About 10,000 food stores and 330,000 employees are part of the Rewe Group in Germany today. Another 5000 food stores in other countries (such as the Billa chain in Austria) belong to the Rewe Group, as well as the German discounter Penny, two DIY (do it yourself) superstore chains, and several travel agencies.

Aldi and the Schwarz group are the two top-ranked private companies in Germany. Aldi actually consists of two separate regional companies, namely Aldi North and Aldi South, that are commonly taken together as Aldi and count a total of 62,000 employees. The Schwarz group has 320,000 employees.

Finally, the Metro group is a joint-stock company with two major private shareholders accounting for roughly 64 % of the shares. The Real consumer markets and the Metro cash-and-carry markets belong to the Metro group.

Table 11.3: Turnover in German food retailing by retail chain in 2017

Retail chain	Total turnover in million Euro	Share of total turnover in %	Food turnover in million Euro	Share of food turnover in %
Edeka	56497	23.3	51650	25.7
Rewe Group	42599	17.6	31837	15.9
Schwarz Group	38610	15.9	31359	15.6
Aldi	29665	12.3	24325	12.1
Metro	13623	5.6	9882	4.9
Total	242129	74.8	200612	74.3

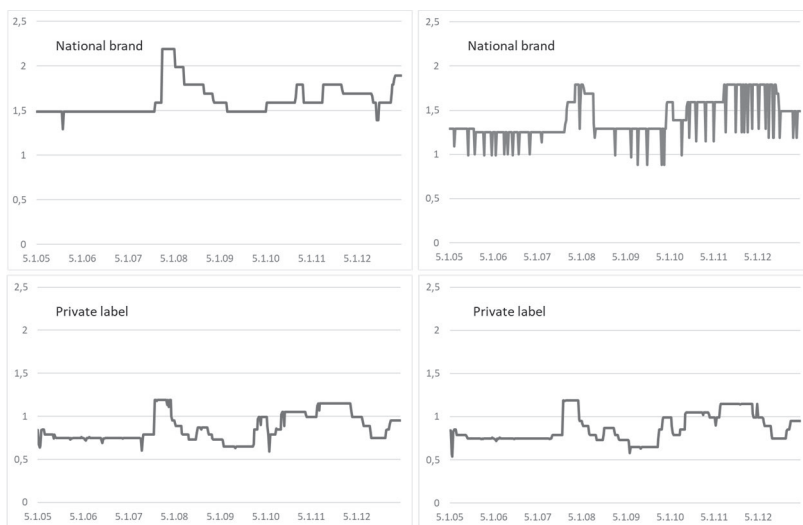
Source: Nielsen 2018b.

11.3.3. Food retail prices

Compared with farm, wholesale, or futures prices, food retail prices exhibit some unique characteristics. Food retail prices in Germany are roughly 6% above the EU average, but relative to above-average incomes in Germany food prices are low. One of the reasons for low relative food prices is that German consumers are particularly price sensitive. For 75% of German consumers the price-quality relationship is the most important measure of a food retailer's performance, followed by quality, product range and convenient location. Price promotions are very important to consumers in choosing products, brands, and stores. The corresponding price strategy based on price promotions or sales is called a **High-Low or HiLo strategy**.

Figure 11.1 shows typical price charts for different items of the same good in different stores over time. It reveals strong price dispersion over time, across stores and brands. We see frequent temporary price reductions (sales) in some cases, and long periods of rigid prices in others. On average, private labels exhibit lower prices and fewer price promotions. Berck et al. (2008) show that it is not profit-maximising for private labels priced close to their marginal costs to offer promotional sales. National brands exhibit strong variation in average prices; some are even priced as low as comparable private labels. Unlike national brands, private labels show significant co-movement (synchronisation) of prices. Private labels often represent the lowest price possible, which is almost the same across stores. Some retailers even explicitly advertise their private label as offering the lowest price on the market.

Figure 11.1: Four examples of weekly retail prices for butter in Germany



Note: Prices in € per 250g.

Source: Own based on IRI Symphony (2014).

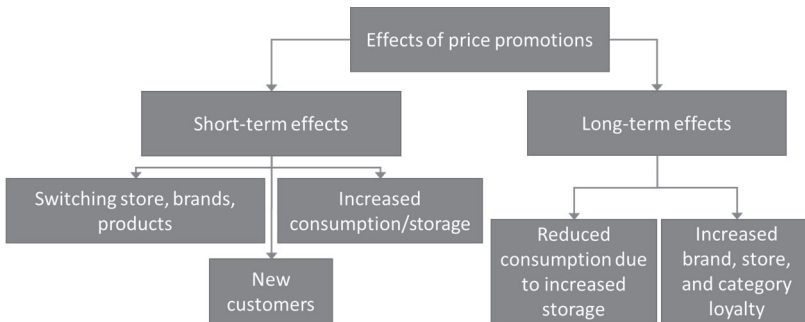
Price promotions

Price promotions are used for various reasons. They might introduce a new product or advertise a product to the consumer. They can be used to clear stocks or to lure customers into a store. In addition, promotions can motivate consumers to switch brands and products. Figure 11.2 summarises the potential effects of promotions.

Promotional sales are a major feature in the marketing mix of many food retailers (Simon 1992, p. 51), accounting for between 20% and 50% of the variation in food retail prices (Hosken and Reiffen 2004, p. 137; Berck et al. 2008, p. 1261). Many models have been proposed to explain sales in an equilibrium price-setting framework, based on different assumptions about retailer motives and consumer behaviour. The **loss-leader** model proposed by Lal and Matutes (1994) assumes advertisement and menu costs. According to this model, retailers only advertise a selection of products. Because consumers have travel costs, they prefer one-stop shopping. Consumers choose a particular store based on the advertised price promotions. Since consumers rationally expect high prices for non-advertised products, stores set some promotional prices even below cost

prices (loss leaders) to attract consumers who then also buy the higher-priced products. An alternative model is Blattberg et al.'s (1981) model of consumer storage. A price promotion gives consumers the incentive to stock up on a storable product, and thus allows the retailer to save on storage costs. In another model, Blattberg and Neslin (1990) show that promotional sales can be the outcome of a prisoner's dilemma. We return to this issue and explain three other models of promotional pricing in more detail in section 11.4.

Figure 11.2: The possible effects of price promotions



Source: Own depiction based on Gedenk (2002, p. 104).

Rigid prices

Menu costs are often used to explain why firms sometimes maintain retail prices at constant nominal levels over several and sometimes lengthy periods of time (see Figure 11.1). Menu costs include all costs related to the adjustment of prices. Food retailers, for instance, often announce their prices by printing price lists, through advertisements, and through in-store labels. Adjusting prices therefore requires printing new lists, placing new advertisements, and changing in-store labels, all of which incurs menu costs. A retailer will only adjust prices if the change in economic conditions, and hence the potential losses from not adjusting prices, exceed these menu costs.³ Dutta et al. (1999) provide evidence that menu costs can be high enough to impede price changes in supermarket and drugstore chains. They estimate that the ratio of menu costs to revenues is 0.70% for supermarket chains, and 0.74% for drugstore chains. Under these conditions, marginal

³ Menus costs are also discussed with respect to asymmetric price transmission in Chapter 10.

changes in the economic environment do not automatically lead to product price changes.

Odd prices and 9-endings

Odd pricing is a strategy whereby a retail price is set just below a round number, for example 9.99 or 8.95. Especially 9-endings are common in food retailing, accounting for 70% to 80% of all prices (Levy et al. 2011; Loy and Weiss 2002). **9-endings** are so-called psychological pricing points that affect consumers' price perceptions. Snir et al. (2017) discuss two underlying mechanisms, the level effect and the image effect. The former occurs when consumers, due to bounded rationality, round prices down or process prices from left to right and ignore the rightmost digit (Stiving and Winer 1997; Thomas and Morwitz 2005). The image effect draws consumers' attention and signals low prices even when alternative products are priced lower (Anderson and Simester 2003). Because of these psychological effects, prices with 9-endings are more rigid (Anderson et al. 2015; McShane et al. 2015).

Price staggering and synchronisation

Rigid prices for a particular firm and product, whatever their underlying cause, do not necessarily imply price rigidity at the aggregate (industry or sectoral) level. At the aggregate level, price rigidity depends on the relative timing of price changes by individual firms. Whether all firms review prices simultaneously (synchronisation) or different firms adjust their prices at different times (staggering) has important macroeconomic consequences. If all firms change their prices simultaneously with a lag of one period following an exogenous shock, the aggregate price level will have adjusted fully to the new equilibrium level by the end of this period and the shock will have no persistent real consequences. If, however, price-setting is staggered, the effect of a nominal shock will be prolonged. In two seminal papers, Taylor (1979, 1980) shows that price staggering can lead to macroeconomic price rigidity. Loy and Weiss (2004) find that price setting is characterised by almost perfect staggering across and within stores in German food retailing.

11.3.4. Regulations

Free competition is a key element of functioning market economies. It stimulates economic performance and offers consumers a broad choice of high-quality products and services at competitive prices. European Union

(EU) competition policy ensures that competition is not distorted in the internal market by ensuring that similar rules apply to all companies operating in the EU. Title VII, chapter 1 of the Treaty on the Functioning of the European Union establishes the legal basis for Community rules on competition. In Germany, the Antitrust Commission is tasked with enforcing the EU competition law.⁴

The laws against unfair competition and against restraints of competition constitute the basis for prosecuting certain business practices. Many forms of aggressive business practices, such as direct threats against competitors or business associates, are forbidden. Comparative advertising is not allowed. Some forms of (price) discrimination, in particular discrimination with respect to religion, ethnicity, or gender, are prohibited. In addition, dominant firms are not allowed to ask for material benefits from their business associates without an objective reason. Permanently setting prices below cost prices is also forbidden. Before 1974, many manufacturers in Germany set fixed resale prices to control the retail prices of their brands. Since 1974 this practice is prohibited, and manufacturers may only make non-binding price recommendations.

11.3.5. Different types of consumers

Consumer expenditures for food, beverages, and **food away from home (FAFH)** in Germany amounted to € 274 billion in 2018, of which € 81 billion was spent on FAFH in restaurants, fast-food restaurant, and canteens etc. The food processing industry has a turnover of € 180 billion and the farm production value is at € 59 billion. Food expenditures altogether account for 11% of total consumer expenditures in Germany.⁵ In Europe, this share ranges from 8% in the UK to 29% in Romania.

Consumers are not homogeneous. Retailers try to take advantage of this variety by offering different products, brands, promotions, and package sizes etc. Marketing activities aimed at individuals have become increasingly prevalent with the advent of internet shopping. Digital marketplaces allow firms to collect and process information about individuals and to process it in real time. This enables them to generate

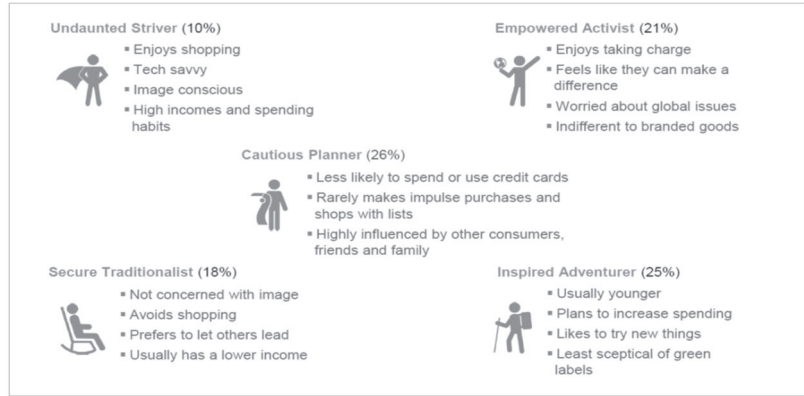
⁴ Section 8 of Chapter 8 (“Pricing under imperfect competition”) discusses EU competition policy in detail.

⁵ At the beginning of the 20th century, the share of food expenditures (including tobacco and alcohol) was about 60%. By 1950 it had dropped to 44% and in 1975 it was 23% (Federal Statistical Office 2020).

individual advertisements, offers, and deals, and thus exploit differences between consumers systematically.

Traditional strategies in food marketing are based on a more general segmentation of consumers. Nielsen, for example, classifies consumers into the following groups: young singles, mid-age singles, old singles, young couples without children, mid-age couples without children, full nest 1, 2 and 3 (households with children at different ages; for example, full nest 1 has one child under 6 years of age), and empty nest with no children. Single households buy less often and spend less money on fast-moving consumer goods. Young singles spend less than elderly singles. The most frequent buyers are mid-age couples whose children no longer live in the household, and old singles (Nielsen 2018b). “By understanding how consumers view themselves, as well as what they prioritise as important, companies can create more tailored strategies, from product development and innovation through to sales and marketing” (Euromonitor International 2020). According to Euromonitor International, German consumers can be classified into five main categories (Figure 11.3): (1) undaunted strivers (10%), (2) empowered activists (21%), (3) cautious planners (26%), (4) secure traditionalists (18%), and (5) inspired adventurers (25%).

Figure 11.3: Future consumer types in Germany



Source: Euromonitor International 2020.

11.3.6. Data sources

Various organisations collect information on consumer behaviour and food retailing. In the following we present four sources of such information in

Germany. These are: consumer panel data provided by the Gesellschaft für Konsumforschung (GfK); retail scanner data from the Information Resources Incorporation Symphony Group (IRI); consumer expenditure data from the Einkommens- und Verbrauchsstichprobe (EVS, translation: Income and Consumption Panel) survey; and price data gathered by scraping websites.

Consumer panel data

The GfK runs household and individual consumer panels to collect reports on food and fast-moving consumer good purchases. These self-reports include prices, volumes, promotion indicators, points of purchase, and dates of purchase, as well as information on the reporting person or household. Depending on the product, panellists report their own purchases or the purchases of the household, generally aggregated on a weekly basis. The GfK is a German marketing research and consulting company, founded in 1934 as a non-profit organisation. It now operates worldwide and is ranked sixth among the top market research companies with an annual revenue of \$US 1,626 million in 2018.⁶

The GfK launched its first household panel in 1957. Today the household panel includes over 30,000 of the 39 million households in Germany. Since about 20% of the panellists leave every year, the GfK has to promote panel participation continuously to find replacements. The panel is organised to be representative of the population in dimensions such as regional origin, age, income, employment and household size. Participation is voluntary and panellists receive no financial benefits. Panellists scan their grocery shopping at home, using a scanner device provided by the GfK. Panellists enter the price of every item, point of purchase and whether a product was on (price) promotion. Data are transferred online to the GfK, where they are validated before they are added to the database. For scanning the GfK uses the **EAN (European Article Number)** code. EAN codes need to be updated for new or modified products, which can make it difficult to analyse longer time periods. While the GfK data cover household fast-moving consumer goods purchases, they do not capture purchases by big consumers, the catering trade, tourists, or households' external food purchases. As consumers do not purchase every product in every week at every store, no

⁶ Another important supplier of consumer panel data in Germany is The Nielsen Company, which also operates worldwide with a head office in the USA (<https://www.nielsen.com/de/>). The Nielsen Group is ranked first with a revenue of over \$US 6,500 million (ESOMAR Global Market Research Report 2019).

continuous time series of prices at retail stores can be constructed from this GfK data set.

Retail panel data

Another source of information is retail scan data. All retailers use electronic check-outs to record the volume and price data of every purchase. Some retailers sell these data to market research companies or make them available to researchers. One of the market research companies working in this field is IRI, located in Düsseldorf. IRI operates worldwide and is the seventh-ranked market research company. It purchases retail scan data, analyses it for customers such as brand manufacturers, and resells it. Retail scan data consists of weekly sale volumes and prices for every retail outlet and every product sold by retailers in the sample. Price promotions are not reported by the retailer. IRI adds a promotion indicator variable for each product item by applying a simple algorithm. If a price is reduced by more than 5% against the regular price and if the reduced price is effective for no more than 4 weeks, then it is considered a promotional price. Hence, any price that is maintained for more than 4 consecutive weeks is considered a regular price.

A major disadvantage of retail scan data is self-selection by retailers; for example, the chains Aldi and Lidl do not sell their scanned data. In addition, specialised food stores such as bakeries or butchers and online food stores are not covered by the sample. However, an advantage of retail scan data is that, for individual products, complete time series of prices can be retrieved for each retail outlet in the sample that carries the product.

Income and Consumption Panel

The German EVS (income and consumption panel) provides official statistics on the living conditions of private households. Among other variables, the consumption expenditures of a representative set of private households are recorded. The data are used to determine consumer price indices, to calculate the income distribution in Germany, to define poverty lines, and to feed into the national accounts. The survey is voluntary and is conducted every five years. The sample is selected according to population parameters and consists of about 60,000 households. All participants report their income and expenditures for three consecutive months. To capture seasonal fluctuations, the total sample is divided into four equal parts that report in consecutive quarters. A detailed monthly record is compiled for foods; however, no brands or other product-specific information is gathered,

and product groups are highly aggregated. The data are available to researchers for a small fee, but the low frequency of data collection and lack of brand information are major disadvantages.

Web scraping

Web scraping is also called web harvesting or web data extraction. Web scraping software can be employed to access the World Wide Web using the Hypertext Transfer Protocol, or through a web browser. Web scraping can be done manually or by automated processes using a bot or web crawler. Specific data are gathered and copied from web sites, usually into a local database or spreadsheet. The programming languages Python and R are typically used to write web scraping code. Regular prices, price promotions, product information, product numbers etc. can be extracted *ad infinitum*. The advantages of these data are their high frequency and the fact that information on promotions is provided by the retailer and does not have to be constructed by the market research company or the researcher (see below). A disadvantage is that no information on regular or promotional sales' volumes is available.

Unless it is clearly forbidden by the online retailers or they take countermeasures, web scraping is possible for every public web site. However, some legal aspects of web scraping are unclear. As long as only publicly available information is scraped and no general terms and conditions of business are signed, web scraping appears to be legal. Generally, web scraping should not extract massive data (considered to be more than 20% of all data available). Data extraction that disturbs the operation of the target website is likely to be illegal. Data gathered for scientific research purposes is probably uncritical.

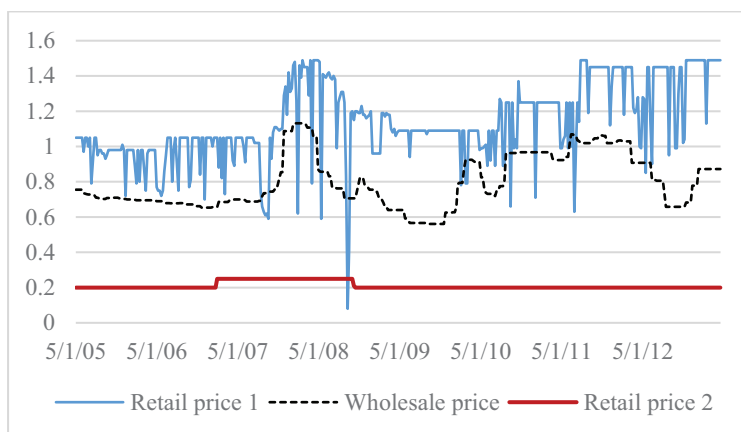
Data quality and discussion

Data quality has two dimensions: data needs to be representative with respect to the research question at hand (validity) and with respect to the underlying population (reliability). We cannot discuss the validity of the four data sources presented above without first defining a specific research question. The research question also determines the relevant underlying population and therefore the reliability of the data. Assume, for example, that the relevant population is all food consumers in Germany. In this context we have already discussed some shortcomings of the consumer scan and retail scanner data: participation in the GfK panel is voluntary and not

all retailers share their retail scan data. Further data shortcomings can arise from measurement and processing errors.

Figure 11.4 shows two time series with potentially inconsistent data. Retail price 1 includes an observation of € 0.08 per 250 g of butter. This price is far below the wholesale price of butter, which is above € 0.5. Perhaps this observation is the result of an entry or processing error. Retail price 2 is below the wholesale price over the entire sample period. Perhaps this retail price is actually for a different package size.

Figure 11.4: Examples of potentially inconsistent retail scan prices of butter



Note: Prices in € per 250g.

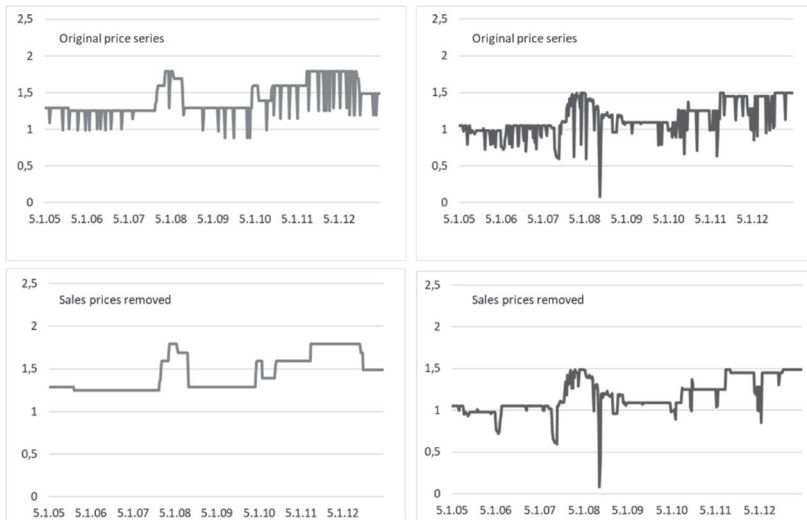
Source: Own based on IRI Symphony (2014).

Another issue is the identification of promotional price sales, which are not indicated by the retailers in scanner data. The algorithm that IRI applies was described above. This algorithm has several obvious problems in identifying price promotions. Holzer and Bittmann (2019), based on Eichenbaum et al. (2011) and Kehoe and Midrigan (2015), developed an algorithm that uses rolling-window modal prices to identify first regular prices and then promotional prices. Figure 11.5 presents two retail price series for butter before (above, original price series) and after (below, sales price removed) the application of this algorithm. In Figure 11.5, each promotional sales price identified by the algorithm in the original series is replaced by the last regular price. For the price on the left, the algorithm works as expected and all obvious temporary price reductions are removed. For the price on the right

right, some observations that cannot be attributed to either promotional or regular prices remain after application of the algorithm.

A variety of other methods for identifying promotions in retail price series have been proposed – see Tifaoui and von Cramon-Taubadel (2016) for an overview.

Figure 11.5: Identification of promotional sales prices in a retail scan price of butter



Source: Own based on IRI Symphony (2014).

While retail scanner data may contain fewer processing errors, consumer scan data rely heavily on entries made by consumers. Prices and promotion indicators have to be manually entered by the panellist. This is a major source of errors, as panel members enter their data at home after returning from shopping. Thus, they need to precisely recall prices and promotions for all items bought. Errors in recollection will lead to noise in the price and promotion information in the data.

11.4. Models and price systems in food retail pricing

As described above, food retailing in Germany is dominated by a few large retail chains. The same is true in many other countries. Only a few large food manufacturers are in a position to deal on equal terms with these retail chains. Thus, retailers can exert market power vis-à-vis food manufacturers.

It is obvious that they can also exert market power vis-à-vis the many consumers.

The traditional starting point for the analysis of market power vis-à-vis consumers is the monopoly model, which we present next in section 11.4.1. Monopolistic competition (section 11.4.2) builds on the monopoly model and introduces the idea of differentiated products. Food retailing is a multiproduct business with several thousand products that are linked by relationships ranging from substitution to complementarity. Retail pricing needs to account for these relationships, which we discuss in section 11.4.3. In section 11.4.4 we look at the retailers' purchases of inputs. The traditional starting point is here the monopsony model. However, the underlying idea that a monopsonist will push down cost prices by cutting demand appears implausible in the case of food retail. We therefore present a model in which the monopsony retailer negotiates both cost prices and volumes.

The oligopoly model best reflects actual retail market structures in many countries. In section 11.4.5 we therefore briefly present basic oligopoly models and an extension of these models under product differentiation, as well as its implications for cost pass-through, which is a key term in the evaluation of vertical markets. In sections 11.4.6 to 11.4.8 we address specific problems and phenomena in retail pricing, such as price discrimination, promotional sales, and brand loyalty.

11.4.1. Monopoly

A monopoly sells to many buyers who have no alternatives to satisfy their demand. The monopoly firm does not take the market price as given, instead it recognises its influence on the market price and will choose the price that maximises its profit. Consumers are confronted with prices when they purchase food, so it is intuitive to think of the monopolist setting the price (p). However, the monopolist can equally maximise profit by choosing the appropriate quantity. In fact, casting the model in terms of profit maximisation with respect to quantity (q) (as we do in the following) is also analytically more convenient.⁷

$$\max_q [p\{q\}q - C\{q\}] \quad (1)$$

⁷ Maximisation with respect to price results an equivalent condition (see Varian 2014, p. 457 and Tirole 1988, p. 66).

$$\frac{\partial p}{\partial q} q + p\{q\} - C'\{q\} = 0 \quad (2)$$

$$\frac{\partial p}{\partial q} \frac{q}{p} p + p = C'\{q\} \quad (3)$$

$$MR \equiv p \left(1 + \frac{1}{\varepsilon^D}\right) = C'\{q\} \equiv MC \quad (4)$$

In food retailing, many costs such as labour, store and equipment are *quasi* fixed. Menu costs are variable to some extent. The most significant variable cost is the purchase (or cost) price of products such as transportation and logistics services. To simplify the model, we assume that the marginal cost (*MC*) equals the purchase price of these services. To maximise profits, the monopolist must equate marginal revenues (*MR*) to *MC*. *MR* depends on the price and the price elasticity of demand for the food product.

The degree of market power can be measured by the **Lerner Index**, which is defined in equation (5). The extent of market power depends directly on the elasticity of demand. If the consumer's reaction is inelastic, the Lerner Index tends towards 1.⁸ If consumers react perfectly elastically, then the Lerner Index, even for a monopolist, is zero.

$$\text{Lerner-Index} \equiv \frac{p-MC}{p} = \frac{1}{|\varepsilon^D|} \quad (5)$$

For linear demand, marginal revenue is also linear. The marginal revenue function intersects the price axis ($q=0$) at the same point as the demand function. The slope of the marginal revenue function is negative and twice that of the demand function. The following derivation proves this.⁹

$$p = a - bq \text{ with } a > b > 0 \quad (6)$$

$$R \equiv pq = aq - bq^2 \quad (7)$$

$$MR \equiv \frac{\partial R}{\partial q} = a - 2bq \quad (8)$$

⁸ The Lerner Index will exceed 1 if the elasticity of demand is less than one, but a profit maximising monopolist would, in the short term, never operate on the price inelastic range of a demand curve.

⁹ The sufficient condition for profit maximisation is that the second derivative of the profit function is negative. This is fulfilled for a linear demand function (Tirole 1988, p. 69).

Since the optimal quantity for a monopolist (q^M) obtains where MR equals MC :

$$q^M = \frac{a-MC}{2b} \quad (9)$$

$$p^M = a - b \left(\frac{a-MC}{2b} \right) \quad (10)$$

$$p^M = \frac{a+MC}{2} \quad (11)$$

Hence, for a linear demand function the monopoly price (p^M) is equal to half of the sum of a plus MC . Furthermore, assuming constant MC , the quantity offered by the monopolist (q^M) is half of the volume that would be offered on a perfectly competitive market. Otherwise, the slope of the MC -curve also determines the equilibrium quantity. The steeper the MC -curve, the smaller the difference between q^M and the perfect competition equilibrium quantity.

In the following section, we briefly analyse cost pass-through (CPT), which is defined as the change in price caused by a change in MC . “Cost pass-through is relevant to an extensive range of competition issues” (Walters et al. 2014, p. 9). We obtain the CPT by deriving the equilibrium price in equation (11) with respect to MC :

$$\frac{\partial p^M}{\partial MC} = \frac{\partial p^M}{\partial a} = \frac{1}{2} \quad (12)$$

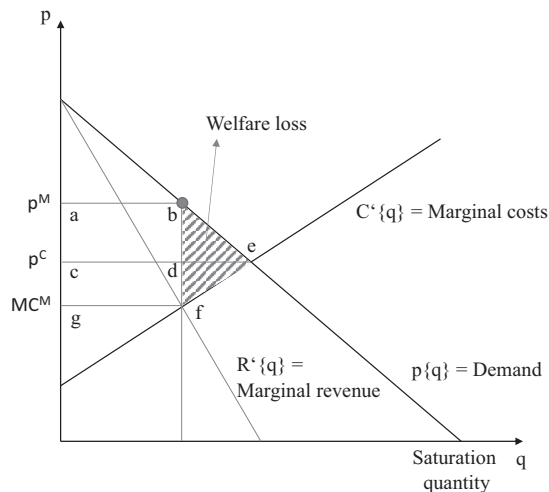
Hence, under monopoly conditions, only 50% of a variation in MC is passed on to consumers. Under perfect competition, the equilibrium condition is $p = MC$, so CPT is one-to-one.

Linear demand functions cause moderate CPT because the demand elasticity increases with price. When the monopolist raises the price, consumer reactions become more elastic and therefore a lower mark-up maximises profits. For isoelastic demand functions, the optimal mark-up for profit maximisation is the same regardless of the cost. For a MC -curve with a positive slope, the partial derivatives do not change; however, for discrete changes the CPT depends on the slopes of the supply and the demand curves, even under perfect competition. In this case, the CPT reference is not one-to-one: the pass-through rate for a monopoly is still lower than under perfect competition, but the difference can be small. For non-linear demand functions, no general statement can be made, and the CPT rate

might even be higher for a monopoly than under perfect competition. For an isoelastic demand and constant marginal costs, for example, a cost change implies a one-to-one pass-through rate for perfect competition, while a monopolist will keep the same relative mark-up, implying that the price in absolute terms increase more than the cost.

Figure 11.6 shows that a monopolist would offer at most half of the saturation quantity (market demand at a price of zero). Therefore, under conditions of linear demand a monopolist will never enter the region of inelastic demand (see also footnote 9). Inelastic demand implies negative marginal revenues, hence any quantities supplied in this region reduce revenues while increasing costs, which is not compatible with profit maximisation. However, this might be short-sighted if consumers react more elastically in the long-term. When the government enforces a minimum supply or when potential rivals enter the market at higher prices, a monopolist might also supply an inelastic (low price) demand.

Figure 11.6: The monopoly market



Compared with the outcome under perfect competition, the monopoly outcome leads to welfare losses equivalent to the area *bef* in Figure 11.6. Consumers lose *abce* of their net surplus, while the monopolist loses producer surplus *def* but gains *abcd* due to the price increase. As the monopolist maximises profits, the net effect must be positive; thus *abcd* > *def*. The net welfare loss for society is *bef* due to reduced production

compared with perfect competition. The net welfare effect may be less negative or even positive if the monopolist has a cost advantage, e. g. due to economies of scale. The difference between the monopoly price and the marginal cost, the area *abfg* in Figure 11.6, is called the monopoly rent.

Monopolistic profits create incentives to develop new products, and patent protection ensures that firms can realise monopoly rents for their inventions. However, inventions in the food industry most often take the form of new recipes, which become known when inventors register patents. Thus, most food processors choose to keep their recipes secret.

11.4.2. Monopolistic competition

The food retail market in Germany is not a monopoly; it is dominated by more than one large retail chain. However, some products or brands might be unique from a consumer perspective, and this might allow the corresponding manufacturers and retailers to exercise monopoly power. It is possible to estimate the extent of this market power by comparing prices between manufacturer (national) and basic retailer (private label) brands. To this end, Connor and Peterson (1992) and Barsky et al. (2003) use retailer brand prices as an indicator of marginal costs to calculate mark-ups on national brands. Low price private labels best represent purchasing prices or production costs. Thus, all mark-ups on national brands compared with low-price private labels can be interpreted as short-run excess profits and an indication of the degree of market power exercised by manufacturers and/or retailers. The extra profit may be divided between manufacturer and retailer, depending on their respective negotiation positions, consumer loyalty towards the brand, and long-term costs.

We illustrate this using fluid milk from the Rewe group's online grocery store as an example. In total, the Rewe online store offers more than 50 different fluid milk items. In Table 11.4 we present data on 30 of these items to illustrate the main factors that influence in-store product price dispersion. Drinking milk is differentiated with respect to the brand (private label, national brands, brand characteristics), the fat content (1.5 to 3.9%), the production process (pasture-fed, organic, etc.), with or without lactose, perishability (pasteurised, extended shelf life, ultra-high temperature), and package size (tetra pack or glass bottle). The lowest-priced private label is "ja!" milk, which sells for 0.79 €/l (3.5% fat). "Weihenstephan" brand milk with 3.5% fat is priced at 1.29 €/l. The price difference between the two brands is 0.50 €/l., which suggests a Lerner Index of roughly 39%.

The most expensive milk is from Demeter, an organic milk with 3.7% fat in a glass bottle at a price of 1.89 Euro/l. Rewe offers the low-price private label “ja!”, as well as several higher-quality labels such as “Rewe Beste Wahl” and “Rewe Bio”, which are priced close to the lowest-price manufacturer brand, which is “Berchtesgadener Land” at a price of 1.15 €/l for 3.5% fat milk.

As long as the entry to the market is free, all manufacturers’ profits should be zero in the long-term. If mark-ups vary as much as the example in Table 11.4 suggests, and if indeed all profits are zero, then additional fixed and variable costs must explain the different mark-ups. The idea of monopolistic competition is that the average cost curve touches the demand curve at the monopoly price to ensure an equilibrium (Figure 11.7). In reality this equilibrium might never be attained. However, it is reasonable to assume that strong brands have higher marketing and advertising costs that account for most or at least some of their mark-ups. In addition, there are significant costs associated with the research, development, and introduction of new brands, many of which fail. These costs have to be borne by those brands that are successful, which can also explain higher mark-ups.

Figure 11.7: Monopolistic competition

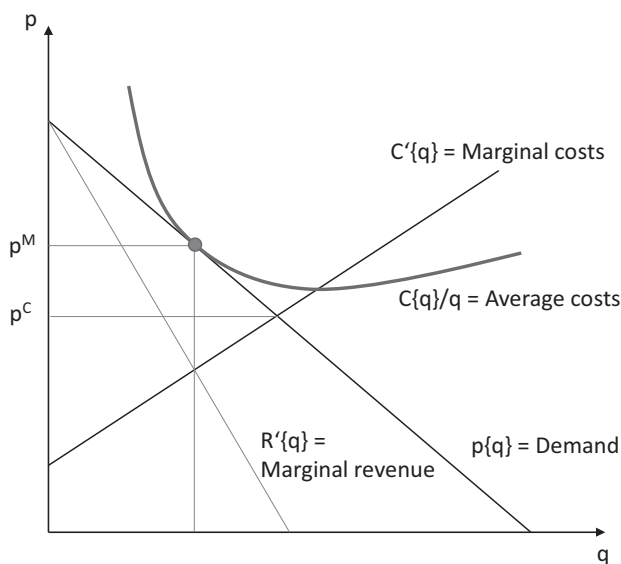


Table 11.4: Drinking milk prices at the Rewe online store on March 3, 2021

Brand	Type	Description	Size	Pack- age	Fat (%)	Lac- tose	Perish- ability	Price (€/l)	Nutri- score
ja!	PL	Skim	1L	TP	1.5	yes	P	0,71	A
ja!	PL	Skim long-life	1L	TP	1.5	yes	UHT	0,71	A
ja!	PL	Whole	1L	TP	3.5	yes	P	0,79	B
ja!	PL	Long-life	1L	TP	3.5	yes	UHT	0,79	B
Hansano	NB	Skim long-life pasture-fed	1.5L	TP	1.5	yes	UHT	0,90	A
Rewe Bio	PL	Organic skim long-life	1L	TP	1.5	yes	UHT	0,99	A
Rewe Beste Wahl	PL	Pasture-fed	1L	TP	3.5	yes	P	1,09	B
Berchtesgadener Land	NB	Fresh, skim alpine farmers	1L	TP	1.5	yes	ESL	1,09	A
Rewe	PL	Long-life, lactose-free	1L	TP	3.5	no	UHT	1,09	B
Rewe Bio	PL	Organic long-life	1L	TP	3.8	yes	UHT	1,09	B
Berchtesgadener Land	NB	Fresh, alpine farmers	1L	TP	3.5	yes	ESL	1,15	B
Bärenmarke	NB	Fresh, alpine farmers	1L	TP	1.8	yes	ESL	1,19	A
Weihenstephan	NB	Fresh, skim alpine farmers	1L	TP	1.5	yes	ESL	1,19	A
Weihenstephan	NB	Fresh, alpine farmers	1L	TP	3.5	yes	ESL	1,29	B
Hansano	NB	Fresh, pasture fed	1L	TP	1.8	yes	ESL	1,29	A
Weihenstephan	NB	Long-life	1L	TP	3.5	yes	UHT	1,29	B
Landliebe	NB	Fresh, country	1.5L	TP	3.8	yes	ESL	1,33	B
Hansano	NB	Fresh, pasture-fed	1L	TP	3.9	yes	ESL	1,35	B
Hamfelder Hof	NB	Organic long-life	1L	TP	3.8	yes	ESL	1,35	B
Hamfelder Hof	NB	Organic skim	1L	TP	1.5	yes	ESL	1,35	A
Hofmolkerei Dehlwes	NB	Organic	1L	TP	3.5	yes	P	1,39	B
Andechser	NB	Organic skim	1L	TP	1.5	yes	ESL	1,39	A
Andechser	NB	Organic	1L	TP	3.8	yes	ESL	1,49	B
Arla	NB	Fresh, skim organic pasture-fed	1L	TP	1.5	yes	ESL	1,49	A
Arla	NB	Fresh, organic pasture-fed	1L	TP	3.8	yes	ESL	1,59	B
Landliebe	NB	Fresh, country	1L	B	3.8	yes	ESL	1,59	B
Landliebe	NB	Lactose-free	1L	TP	3.8	no	P	1,65	B
Hansano	NB	Fresh, pasture-fed	0.5L	TP	3.9	yes	ESL	1,7	B
Hansano	NB	Fresh, skim lactose-free	1L	TP	1.5	no	P	1,75	A
Demeter	NB	Organic	1L	B	3.7	yes	P	1,89	B

Notes: PL = private label; NB = national brand; TP = tetrapack; B = bottle; P = pasteurised; ESL = extended shelf life; UHT = ultra-high temperature. Source: Own depiction based on <https://shop.rewe.de/>, accessed on March 3, 2021.

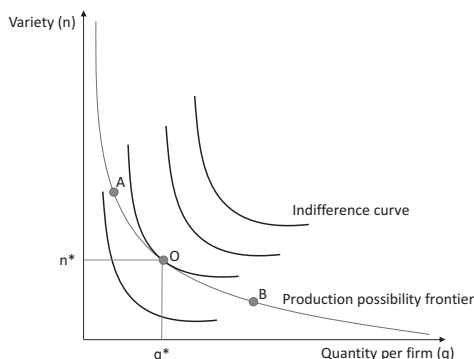
How many and how differentiated should products be produced under monopolistic competition? Carlton and Perloff (2015, p. 244) describe the optimal solution graphically as depicted in Figure 11.8. First, the production possibility frontier indicates that there is a trade-off between producing variety and quantity; firms can produce greater variety but at the expense of quantity. Second, the convex social indifference curves indicate that the marginal benefits of increasing variety and increasing quantity, respectively, are decreasing. The social optimum is reached at point O , where a societal indifference curve is tangent to the production possibility frontier, n^* different products are produced and the output quantity is q^* .¹⁰

Monopolistic competition equilibrium can lead to equilibria with either too much variety (such as A) or too little (such as B) relative to the social optimum (Carlton and Perloff 2015, p. 243). While fixed costs and economies of scale lead firms to produce too few brands, substitution between products increases a firm's profits by reducing the demand for the products of competing firms. This results in firms offering too many brands.

So far, we have implicitly assumed a representative consumer, for whom all brands are equally good substitutes. In reality, however, it is rarely the case that consumers view different brands as equally good substitutes. Furthermore, individual consumers differ in their perceptions of brands and brand substitutions. To overcome this limitation, Hotelling (1929) introduced a location model of product differentiation. Salop (1979) extended the linear model to a circular model. The difference between products is represented by the geographical distance between firms' or products' locations. The closer products are in geographical or characteristic space, the better substitutes they are. Depending on their preferences, consumers also have locations in geographic or product space.

¹⁰ Carlton and Perloff (2015, p. 259 ff.) argue that if products are homogeneous, only one firm should produce at the level where marginal cost equals price. To prevent shutdown, the firm's losses should be levelled by a subsidy.

Figure 11.8: Optimal allocation of quantity and variety of products



Source: Own based on Carlton and Perloff (2015, p. 244).

11.4.3. Multiproduct monopoly

Food retailing is a multiproduct business. Models of product differentiation, such as the monopolistic competition model, are a first step when considering multiple products and the relationships between them. We now introduce multiple products to a monopoly model to extend the analysis.

If a firm produces n products, it has to consider the relationships between them in consumption and production. If the products were independent in production and consumption, each could be analysed using a standard single-product model. Accounting for interdependence requires more complex models. In this section, we focus on relationships between products in consumption and assume that the products are independent in production. Niehans (1956) investigated this setting and presented a solution, sometimes referred to as the ‘Niehans-Rule’. A very similar result was published by Ramsey (1927) in his theory of optimal taxation. Thus, the resulting prices are also called Ramsey prices.

Products can be either substitutes or complements in consumption, depending on consumer preferences. Products can be chemically or physically different, but nevertheless viewed as substitutes by consumers. Conversely, products can be identical but nevertheless be viewed as different by consumers. Some consumers may perceive cola and lemonade as substitutes, others might prefer “mezzo mix” and thus perceive cola and lemonade as complements. “The consumer is always right” (Carlton and Perloff 2015, p. 227).

To keep the algebra simple, we assume two products and the following profit function:

$$\max_{p_1} [p_1 q_1 \{p_1, p_2\} + p_2 q_2 \{p_1, p_2\} - C\{q_1 \{p_1, p_2\}, q_2 \{p_1, p_2\}\}] \quad (13)$$

We differentiate equation (13) with respect to the price of product 1 (p_1), and set the derivative equal to zero. After a few modifications, we obtain the optimal $p_1(q_1^*)$:

$$q_1 \{p_1, p_2\} + p_1 \frac{\partial q_1}{\partial p_1} + p_2 \frac{\partial q_2}{\partial p_1} - \frac{\partial C}{\partial q_1} \frac{\partial q_1}{\partial p_1} - \frac{\partial C}{\partial q_2} \frac{\partial q_2}{\partial p_1} = 0 \quad (14)$$

$$q_1 \{p_1, p_2\} + \frac{\partial q_1}{\partial p_1} \left(p_1 - \frac{\partial C}{\partial q_1} \right) + \frac{\partial q_2}{\partial p_1} \left(p_2 - \frac{\partial C}{\partial q_2} \right) = 0 \quad (15)$$

$$p_1 + \frac{\partial q_1}{\partial p_1} \frac{p_1}{q_1} \left(p_1 - \frac{\partial C}{\partial q_1} \right) + \frac{\partial q_2}{\partial p_1} \frac{p_2}{q_2} \left(p_2 - \frac{\partial C}{\partial q_2} \right) = 0 \quad (16)$$

$$p_1 + \varepsilon_{11} \left(p_1 - \frac{\partial C}{\partial q_1} \right) + \varepsilon_{21} \frac{q_2}{q_1} \left(p_2 - \frac{\partial C}{\partial q_2} \right) = 0 \quad (17)$$

$$p_1^* = \frac{\varepsilon_{11}}{1 + \varepsilon_{11}} \frac{\partial C}{\partial q_1} - \frac{\varepsilon_{21}}{1 + \varepsilon_{11}} \frac{q_2}{q_1} \left(p_2 - \frac{\partial C}{\partial q_2} \right) \quad (18)$$

In these equations, ε_{11} and ε_{12} are the elasticities of demand for product 1 with respect to the prices of product 1 and product 2, respectively. The first term of the right-hand side of equation (18) is the standard profit maximising price for a single-product monopoly. The sign of the second term depends on the cross-price elasticity ε_{12} . For substitutes, $\varepsilon_{12} > 0$, and for complements $\varepsilon_{12} < 0$. Note that $1 + \varepsilon_{11} < 0$, as monopolists only supply in the price-elastic range of the demand curve. Moreover, the quantities q_1 and q_2 and the mark-up $\left(p_2 - \frac{\partial C}{\partial q_2} \right)$ must be greater than zero.

Thus, the second term is positive for substitutes $\left(-\frac{\varepsilon_{21}}{1 + \varepsilon_{11}} \frac{q_2}{q_1} \left(p_2 - \frac{\partial C}{\partial q_2} \right) > 0 \right)$. This implies that the profit maximising p_1 will be higher than the standard single-product monopoly price if the monopolist is also selling a substitute product 2. The additional price mark-up increases with the substitutability of the products, the magnitudes of ε_{11} and ε_{12} , the ratio of quantities $\frac{q_2}{q_1}$, and the mark-up for product 2.

This result makes intuitive sense. For independent products, profits are maximised when marginal revenue equals marginal cost. For substitutional

products, the monopolist has to consider the additional demand for product 2 that is caused by an increase in p_1 . This additional demand increases the marginal revenue by increasing p_1 . For complements, the sign of the second term on the right-hand side of equation (18) is negative, implying that the profit maximising p_1 will be lower than in the independent-product monopoly case.

A similar interpretation follows from the profit maximising Lerner Index in equation (19). For the general case with n products, we can similarly derive the profit maximising price and the Lerner Index (equations 11.20 and 11.21). The price effect depends on the number and nature (substitutes, complements) of the product relationships in the monopolist's assortment.

$$\frac{p_1^* \frac{\partial C}{\partial q_1}}{p_1^*} = -\frac{1}{\varepsilon_{11}} - \frac{\varepsilon_{21}}{\varepsilon_{11}} \frac{q_2}{p_1 q_1} \left(p_2 - \frac{\partial C}{\partial q_2} \right) \quad (19)$$

$$p_i^* = \frac{\varepsilon_{ii}}{1 + \varepsilon_{ii}} \frac{\partial C}{\partial q_i} - \sum_{j \neq i} \frac{\varepsilon_{ji}}{1 + \varepsilon_{ii}} \frac{q_j}{q_i} \left(p_j - \frac{\partial C}{\partial q_j} \right) \quad (20)$$

$$\frac{p_i^* \frac{\partial C}{\partial q_i}}{p_i^*} = -\frac{1}{\varepsilon_{ii}} - \sum_{j \neq i} \frac{\varepsilon_{ji}}{\varepsilon_{ii}} \frac{q_j}{p_i q_i} \left(p_j - \frac{\partial C}{\partial q_j} \right) \quad (21)$$

Applications of the Niehans-Rule are rare - even in scientific journals - for the following reasons. First, it requires estimates of own-price and cross-price demand elasticities for all products. Even in a small supermarket, the number of products can easily exceed 5000. Although it can safely be assumed that $\varepsilon_{ij} = 0$ for many product pairs, many thousand elasticities remain to be estimated. Estimation is further complicated by the fact that many prices are highly collinear over time, which inflates standard errors, and by price promotions, as it is almost impossible to disentangle their various effects such as increased storage, brand switching, etc.

11.4.4. Monopsony

Average retail chain profits are lower in Germany than in other European countries. Relative to income levels, retail prices in Germany are also comparatively low. This suggests that German retailers are exercising little or no market power as sellers to consumers. Nonetheless, the market concentration ratio is very high. Therefore, the German Antitrust Commission has investigated in the market power of food retailers vis-à-vis the manufacturing or food-processing industry.

Analogous to section 4.1 above, where we considered retailers' market power as sellers to consumers, we now begin the discussion of retailers' market power as buyers, with the monopsony model for one product and one input. The main input factor for food retailers, however, is the same as their output good; retailers do not produce goods but rather provide retail services to re-sell food products that they have previously purchased. In this process, product loss (quality and/or quantity) can take place. In addition, retailing involves short-term fixed costs (C), predominantly for labour and capital. Assume a single retailer with given (competitive) output prices, many input (product) suppliers, and the following profit function:

$$\max_x [pq\{x\} - r\{x\}x - C] \quad (22)$$

where r is the purchase price of the input x . We differentiate equation (22) with respect to the input x and set the first derivative equal to zero. After a few modifications, we obtain the profit maximisation condition that the gross value added (GVA) – here, the retail sale price – is equal to the marginal expenditure (ME) of the monopsonist for the input x :

$$p \frac{\partial q}{\partial x} - \frac{\partial r}{\partial x} x - r\{x\} = 0 \quad (23)$$

$$p \frac{\partial q}{\partial x} = \frac{\partial r}{\partial x} \frac{x}{r} r + r\{x\} \quad (24)$$

Assuming that $\frac{\partial q}{\partial x} = 1$ (one unit of input x is resold as one unit of output q), and rearranging:

$$GVA \equiv p = r \left(1 + \frac{1}{\varepsilon^s} \right) \equiv ME \quad (25)$$

$$\text{Lerner-Index}^* \equiv \frac{GVA - r^M}{r^M} = \frac{1}{\varepsilon^s} \quad (26)$$

where ε^s is the price elasticity of supply of the input x . If the input supply is linear, the marginal expenditure function is also linear and intersects the price axis (r given $x = 0$) at the same point as the input supply function. The slope of the marginal expenditure function is twice the slope of the input supply function.

$$r = h + kx \quad (27)$$

$$rx = hx + kx^2 \quad (28)$$

$$ME \equiv \frac{\partial rx}{\partial x} = h + 2kx \quad (29)$$

Since the monopsonist's optimal purchase quantity x^M obtains where $ME = GVA$:

$$GVA = h + 2kx \quad (30)$$

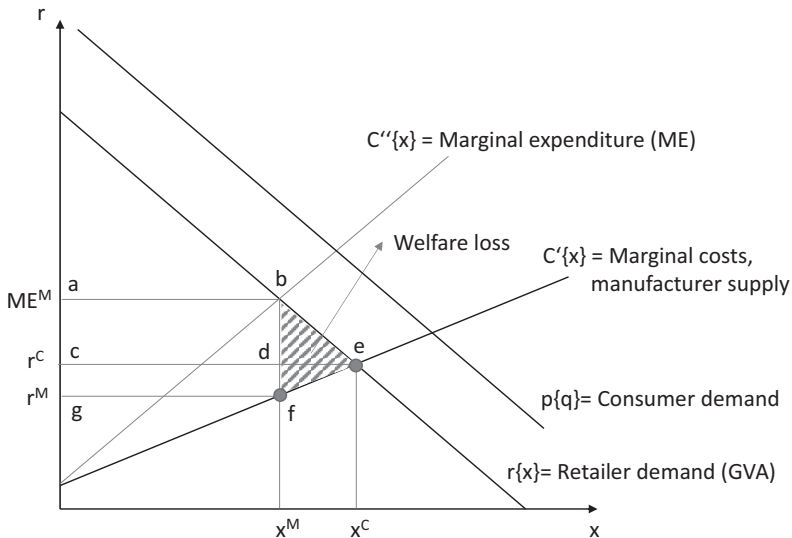
$$x^M = \frac{GVA - h}{2k} \quad (31)$$

$$r^M = h + k \left(\frac{GVA - h}{2k} \right) \quad (32)$$

$$r^M = \frac{h + GVA}{2} \quad (33)$$

Hence, the monopsonist will purchase half as much x as would be purchased under perfect competition, if consumer demand for the output q is perfectly price elastic so that $P = GVA$ is fixed. Otherwise, the slope of the consumer demand function determines the market volume (Figure 11.9).

Figure 11.9: Monopsony market



As in the monopoly case, the steeper the demand, the closer the equilibrium quantity x^M is to the volume under perfect competition x^C . Differentiation of equation (33) with respect to GVA and with respect to c provides information similar to CPT rates:

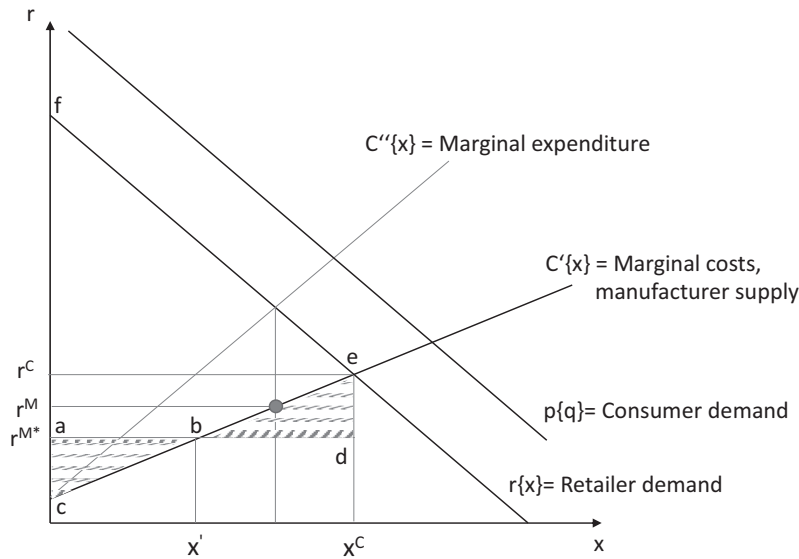
$$\frac{\partial r^M}{\partial h} = \frac{\partial r^M}{\partial GVA} = \frac{1}{2} \quad (34)$$

Both derivatives equal $\frac{1}{2}$. GVA changes when either retail prices or variable retailing costs change. Thus, if these increase by one unit, the purchase price of the input x increases by $\frac{1}{2}$. The same applies for factors that shift manufacturers' costs (for example when the prices of agricultural raw products change). As before, these results assume a perfectly elastic GVA (demand) function. For falling GVA , the effects of manufacturer cost changes on r can exceed $\frac{1}{2}$, and the effects of changes in GVA on r can be smaller than $\frac{1}{2}$.

Haucap et al. (2015) state that this standard monopsonistic model is not capable of illustrating the exploitation of market power by food retailers. Instead of reducing demand in order to push down purchase prices, food retailers negotiate delivery conditions. If we assume that retailers can negotiate prices and delivery volumes, the equilibrium changes. If the retailer has perfect market power, the food manufacturers will have to forego all profit and transfer it to the retailer. The retailer will negotiate the same quantity as under perfect competition at a price that extracts the entire gross margin from the manufacturer(s). This outcome is depicted in Figure 11.10, where the retailer chooses the perfect competition volume x^C but uses market power to push the purchase price down to r^{M*} . As a result, the manufacturers' profits on the first x' units (abc) exactly compensate their losses on the units between x' and x^C (bde).

Even though this is an extreme outcome that might not occur in practice, the idea remains that retailers can use market power to push purchase prices down without reducing the quantity demanded. If retailers pass on reduced purchase prices to consumers, they will need to negotiate even higher volumes to satisfy the increase in consumer demand.

Figure 11.10: Market outcome under buyer market power, negotiating price and volume



In a sector study, the German Antitrust Commission (2014) clearly states that the four leading retail chains in Germany (Edeka, Rewe, Schwarz Group, and Aldi) dominate the competition in the market, in terms of supply and input demand. These four retail chains use negotiations with food processors to their advantage, which causes further concentration in food retailing and the food processing industry. However, concentration in food processing does not necessarily improve the processors' negotiating position, as large food processors have no or fewer fall-back options besides the four leading retail chains. Thus, large food processors may have less power in negotiations with the retail chains than smaller processors. Only a few large food processors that produce 'must have' brands negotiate on an equal footing with the big food retail chains (German Antitrust Commission 2014).

High market shares of private labels support the superior position of food retail chains in negotiations with food processors. However, a food processor that produces a national brand as well as a private label will be better off than a pure national brand manufacturer, if the retailer and

processor can agree on a consistent price differentiation strategy for these brands.

Smaller retail chains in Germany organise purchasing cooperatives, which enable them to generate market power comparable to that of the leading retail chains. Some of these purchasing cooperatives are led by one of the ‘big four’. The additional profit generated by these cooperatives might be small for the big retail chain, but it allows the chain to offer private labels in the cooperatives’ stores, which might ease future mergers, and increase the popularity of the chain.

11.4.5. Oligopoly

An oligopoly is a market with few suppliers or producers and a large number of consumers. This market structure almost perfectly describes the retail food markets in industrialised countries such as Germany.

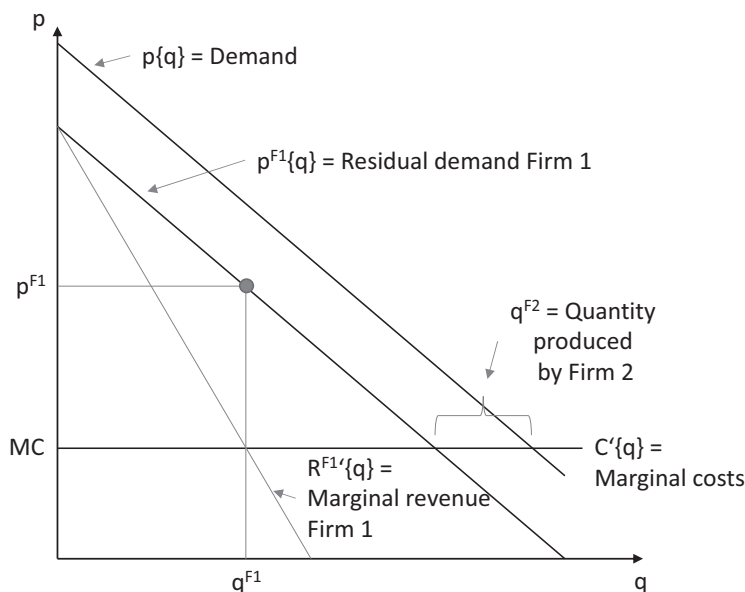
Oligopoly models can be classified into cooperative and non-cooperative models. Cooperative models refer to cartels, the members of which coordinate some or all of their relevant production and marketing decisions. Ideally a cooperative oligopoly behaves as one firm, i.e. a monopoly. The coordination of the cartel group’s actions is easier, when (a) the product is homogeneous, (b) there are few group members, (c) group members have the same production costs, (d) group members are of the same size, and (e) antitrust laws are weak. Under German antitrust laws, cartels are illegal and the German Antitrust Commission actively investigates suspected cases. Examples are reported on the website of the German Antitrust Commission at <https://www.bundeskartellamt.de>.

In non-cooperative oligopolies, the competitors’ reactions to each other’s actions are essential. In the following section, we briefly describe the three traditional approaches to modelling competition among oligopolists. A more comprehensive presentation can be found in Chapter 8 (“Pricing under imperfect competition”). After presenting these models, we discuss an oligopolistic model to investigate the impact of product differentiation on CPT. This provides the theoretical background to an empirical case study presented in section 5.4.

Augustin Cournot developed an oligopoly model in 1838. In the simplest variant he assumed two firms, a duopoly, which produce a homogeneous product. Both firms have the same production function and the same constant marginal costs of production. The strategic variable is the quantity

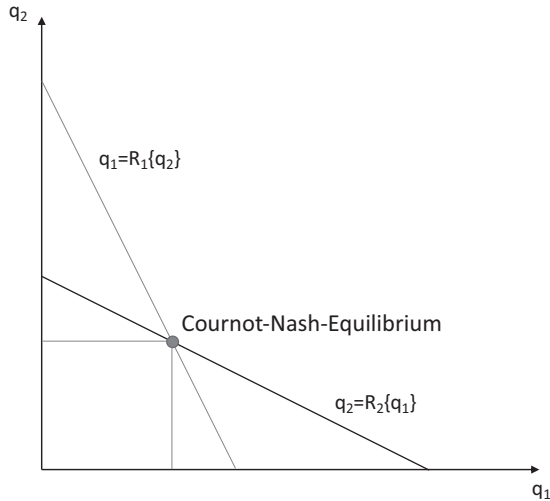
produced. Cournot assumes that once a firm (firm 2) has decided the quantity of its production (q^{F2}), it does not react to the competitor's (firm 1) response. Thus, firm 1 takes the firm 2's production as given. In this case the firm 1's residual demand equals the total demand minus firm 2's production, as depicted in Figure 11.11. Firm 1 applies monopolistic behaviour to its residual demand function and thus chooses to produce q^{F1} . For any q^{F2} chosen by firm 2, firm 1 can calculate its optimal q^{F1} , and vice versa. Based on these considerations, each firm can calculate its own and its competitor's **reaction function**. These reaction functions are depicted in Figure 11.12.

Figure 11.11: Residual demand in the Cournot duopoly



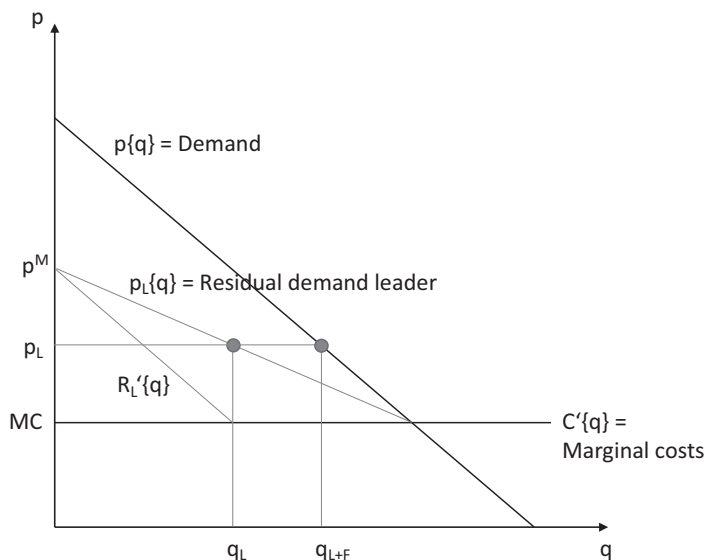
The intersection of the reaction functions marks the **Cournot-Nash-Equilibrium**. Following Carlton and Perloff (2015, p. 185) “a set of strategies is called a Nash-Equilibrium if, holding the strategies of all other firms constant, no firm can obtain a higher payoff (profit) by choosing a different strategy”. The duopoly model can be extended to n firms, and the market equilibrium demonstrably converges to the competitive equilibrium as n approaches infinity. For $n = 1$, the monopoly price equilibrium results. Thus, competition increases monotonically with n .

Figure 11.12: Reaction functions in the Cournot duopoly



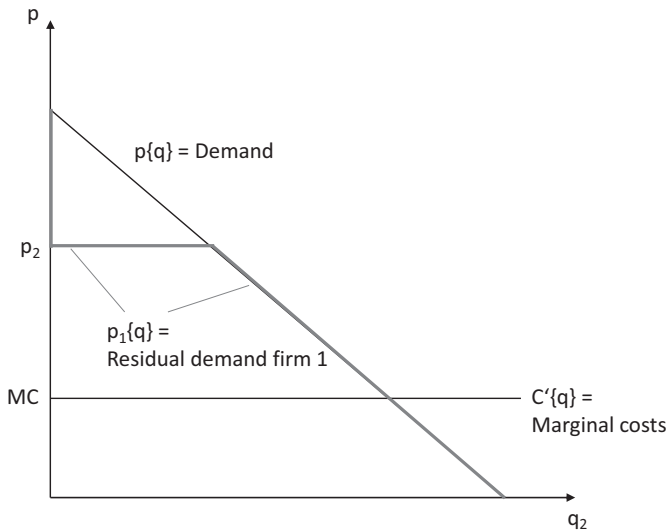
In the Cournot model, the quantity is the strategic variable and actions take place simultaneously. Stackelberg (1934) also developed a duopoly model in which quantity is the strategic variable, but actions take place sequentially, starting with the market leader (L). The follower (F) observes the leader's choice and reacts to it. The leader assumes that the follower will react as a competitor in the Cournot model would, and anticipates this reaction when deriving his/her own residual demand function. For the linear case the leader's residual demand starts at the monopoly price and ends at the equilibrium under perfect competition (Figure 11.13). When the leader supplies nothing, the follower realises the monopoly price equilibrium (p^M). When the leader supplies as under perfect competition, then no demand is left for the follower. The equilibrium price in the Stackelberg duopoly is lower than the Cournot equilibrium price, but the market leader realises a higher profit than the firms in the Cournot duopoly.

In both the Cournot and Stackelberg models, the oligopolists choose quantities. It might be more realistic to assume that the price is the strategic variable on food markets. Therefore, in 1883 Bertrand developed a model of two firms that simultaneously set prices rather than quantities. What does the residual demand function of firm 1 look like, when firm 2 sets a price in the range between the consumers' maximum willingness to pay and the competitive price?

Figure 11.13: The Stackelberg leader (L) follower (F) model

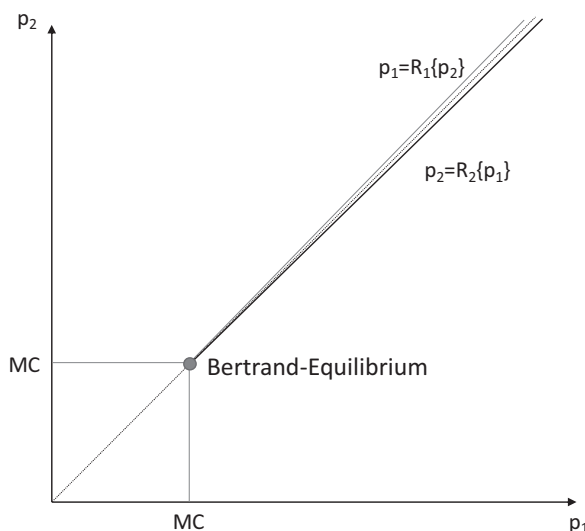
The answer to this question is depicted in Figure 11.14. When firm 2 sets price p_2 , firm 1 can underbid this price and receive the total demand. Thus, as long as p_2 is above the competitive price, which equals the constant marginal cost of production MC , firm 1 will underbid it to capture the entire market demand. Hence, as shown in Figure 11.15, the price reaction functions lie closely above and below the 45-degree straight line, starting at MC , where both functions intersect. Thus, the Bertrand-equilibrium is the same as the competitive equilibrium. This is an astonishing result: for a monopoly or monopsony the choice of the strategic variable is irrelevant to the outcome, but for a duopoly it makes a substantial difference.

Figure 11.14: Residual demand in the Bertrand model



In section 4.2 on monopolistic competition, we introduced the term **product differentiation** without providing a clear definition. Product differentiation in a loose sense may be defined as “any set of commodities closely related in consumption and/or production” (Eaton and Lipsey 1989). Product differentiation has a horizontal and a vertical dimension. If consumers rank products at the same price differently, product differentiation is said to be horizontal. If consumers similarly rank products at the same price, product differentiation is called vertical (Beath and Katsoulacos 1991). Product differentiation is an important feature, not only in food markets. In many industries we find a large number of similar but differentiated products. “There must hardly be a reader who is not aware of the daunting array of breakfast cereals, detergents, toiletries on the average supermarket shelf” (Beath and Katsoulacos 1991 p. 7). Almost no processor produces identical products, and every processor produces a range of differentiated products. Individual consumers, however, only consume a small subset of all available products (Eaton and Lipsey 1989).

Figure 11.15: Price reaction functions for the Bertrand model



In the following we present a Cournot and a Bertrand oligopoly with product differentiation, employing the representative consumer approach. We follow Shubik and Levitan (1980), Zimmermann and Carlson (2010), and Wang and Zhao (2007), and start with a market of i symmetric non-cooperative competitors offering n differentiated products, where $\phi \in [-1, 0]$ reflects the degree of product differentiation. Zero indicates the maximum level of product differentiation, at which products are completely independent. When all products are perfect substitutes, $\phi = -1$.¹¹ Hence, an increase in ϕ represents an increase in average product differentiation or a corresponding reduction in substitutability. q_i is firm i 's output. Equation (35) shows an inverse firm-specific linear demand function (Zimmerman and Carlson 2010).¹²

¹¹ The definition of product differentiation in Zimmerman and Carlson's (2010) model is reversed ($-\phi \in [0, 1]$, where 1 indicates perfect substitution). For ease of comparison, we use $\phi \in [-1, 0]$ throughout.

¹² The assumption of linear demand is important in this setting. As Weyl and Fabinger (2013) show, for non-linear demand functions and market power, CPT rates may be higher than under perfect competition.

$$p_i = \alpha - q_i + \phi \sum_{j \neq i} q_j \quad (35)$$

We derive the profit maximising quantity of firm i under a symmetric Cournot-Nash Equilibrium (sCN), where q_j is a vector of outputs excluding firm i , by solving the reaction function for marginal costs c , assuming symmetric firms. This leads to the following optimal output and equilibrium price-cost relationships:

$$q_i^{sCN} = \frac{\alpha - c}{2 - \phi(n-1)} \quad (36)$$

$$p_i^{sCN} = \frac{\alpha + (1 - \phi(n-1))c}{2 - \phi(n-1)} \quad (37)$$

The resulting CPT under Cournot-Nash price-cost equilibrium is positive, yet smaller than one:

$$\frac{\partial p_i^{sCN}}{\partial c} = \frac{1 - \phi(n-1)}{2 - \phi(n-1)} > 0 \quad (38)$$

For a given number of firms, perfectly differentiated products ($\phi = 0$) produce a pass-through rate of 50% and each individual firm engages in monopolistic pricing for its product. This result is similar to the monopoly case. When all competitors offer homogeneous products ($\phi = -1$), the right-hand side of equation (38) reduces to $(n/n+1)$, CPT depends on the number of firms, and approaches 1 as $n \rightarrow \infty$.

To investigate the impact of product differentiation on firms' equilibrium prices, we differentiate equation (37) with respect to the product differentiation parameter ϕ :

$$\frac{\partial p_i^{sCN}}{\partial \phi} = \frac{(n-1)(\alpha - c)}{(2 - \phi(n-1))^2} > 0 \quad (39)$$

In equation (39), prices increase with product differentiation. To determine the effect of product differentiation on the degree of CPT, we partially differentiate equation (38) with respect to ϕ :

$$\frac{\partial^2 p_i^{sCN}}{\partial c \partial \phi} = -\frac{(n-1)}{(2 - \phi(n-1))^2} < 0 \quad (40)$$

Product differentiation reduces the CPT, or in other words, closer substitutive relationships between products suggest a higher CPT.

Zimmermann and Carlson's (2010) analysis focuses on the role of competition, namely the number of firms in the market. CPT increases with the number of firms and converges to complete (one to one) CPT under perfect competition ($n \rightarrow \infty$). Therefore, product differentiation and a declining number of firms both reduce CPT by increasing the sellers' market power.

So far, we have assumed symmetric firms, but in a differentiated product market this appears to be rather restrictive. Firms have different product portfolios and different firm-specific marginal costs. Retail purchase prices can therefore differ not only due to product differentiation but also due to the cost structures of the food processors and the volumes that they produce.

Wang and Zhao (2007) extend Shubik and Levitan's (1980) model to introduce firm asymmetry. Equation (41) shows the non-symmetric Cournot-Nash (*nsCN*) price-cost equilibrium relationship that they derive (Wang and Zhao 2007, p. 178). Again, for ease of interpretation we transform the product differentiation measure, which ranges from minus infinity to zero ($\phi \in [-\infty, 0]$) (see footnote 11). However, the interpretation of ϕ remains unchanged as higher parameter values indicate more differentiated products.

$$p_i^{nsCN} = \frac{(n-\phi)\alpha}{2n-(n+1)\phi} - \frac{n\phi\bar{c}(n-\phi)}{(2n-\phi)(2n-(n+1)\phi)} + \frac{nc_i}{2n-\phi} \quad (41)$$

where $c_i, \bar{c} \equiv \sum \frac{c_i}{n}$ are firm-specific and average marginal costs, respectively. To determine the effect of changes in firm-specific marginal costs on CPT, we differentiate equation (41) with respect to c_i .

$$\frac{\partial p_i^{nsCN}}{\partial c_i} = \frac{2n^2 - n(n+2)\phi + \phi^2}{(2n-\phi)(2n-(n+1)\phi)} > 0 \quad (42)$$

As for the case of symmetric firms, when $n = 1$ (monopoly) and products are independent ($\phi = 0$), the rate of CPT is $\frac{1}{2}$. As the number of firms increases, CPT remains at $\frac{1}{2}$, regardless of the level of product differentiation. Zimmermann and Carlson (2010, p. 11) simulate the CPT for different numbers of asymmetric firms. For a single firm (monopoly), the CPT rate is $\frac{1}{2}$ and declines with an increasing number of firms. After reaching a minimum, the rate increases and converges back to the monopoly CPT of $\frac{1}{2}$. Firms have little incentive to adjust prices to changes in firm-specific costs because if they do, consumers will quickly switch to other firms that have not experienced cost changes. Hence, firm-specific costs are

passed on at lower rates than for the monopoly case. This effect becomes stronger with an increasing number of firms and with a lower degree of product differentiation. The number of firms, however, also flattens the reaction function in the Cournot model. This leads to a more uniform firm behaviour and to an increasing CPT rate as the number of firms increases until the monopolistic CPT rate is reached (Zimmerman and Carlson 2010).

To derive the impact of product differentiation on firms' equilibrium prices, we differentiate equation (41) with respect to product differentiation ϕ . The result shows that firms increase prices of more differentiated products:

$$\frac{\partial p_i^{nsCN}}{\partial \phi} = \frac{(n-1)n(4n^2\alpha + n(c(\phi-4))\phi + (2c+\alpha)\phi^2 - 4\alpha)}{(\phi-2n)^2(n(\phi-2)+\phi)^2} > 0, \text{ for } n > 1 \quad (43)$$

In equation (44) we differentiate the CPT from equation (42) with respect to the measure of product differentiation (ϕ). Contrary to the finding for the symmetric case, firms offering more differentiated products increase their CPT. The upper limit for the firm-specific CPT is the monopoly rate of $\frac{1}{2}$; the lower limit is zero.

$$\frac{\partial^2 p_i^{nsCN}}{\partial c_i \partial \phi} = \frac{(n-1)n\phi(n(\phi-4)+2\phi)}{(\phi-2n)^2(n(\phi-2)+\phi)^2} > 0, \text{ for } n > 1 \quad (44)$$

To assess the robustness of the Cournot-Nash model under asymmetric cost structures, Zimmermann and Carlson (2010) also derive the Bertrand price-cost equilibrium relationship. They confirm the Cournot-Nash results. The individual CPT rates have the same upper limit. Under competition, CPT rates are always higher, and the rates monotonically decrease with the number of firms. The result with respect to product differentiation is the same: CPT of firm-specific costs increases for more differentiated products. Contrary to the Cournot-Nash model outcome, the presence of product differentiation reinforces the effect of intensified competition; an increase in ϕ positively affects a firm's response to rivals' output decisions. Consequently, CPT rates strictly decline with the number of competing firms (Zimmerman and Carlson 2010).

In conclusion, *ceteris paribus* we expect the prices of more differentiated products to be higher, and CPT of industry-wide cost shocks to be lower: CPT decreases with product differentiation. For homogeneous products and a large (infinite) number of firms, the CPT rate is 1. As in the monopolistic case, the minimum CPT rate of industry-wide cost shocks is $\frac{1}{2}$, implying that 50% of a cost shock is passed on to the retail price. For individual cost

shocks, CPT rates increase with product differentiation, starting at a rate of zero for homogeneous products. Independent (perfectly differentiated) products show a CPT rate of $\frac{1}{2}$ for firm-specific cost shocks. For homogeneous products, individual costs cannot be passed on to consumers as demand reacts perfectly elastically. For differentiated products, firms can pass on costs. For independent products, CPT rates converge to the monopolistic rate of $\frac{1}{2}$.

The theoretical models discussed here do not make any predictions about the dynamics of the CPT process. Borenstein and Shepard (2002) argue that market power slows down CPT. Baker and Bresnahan (2008) point out that product differentiation is a means of generating market power. Thus, CPT will become more sluggish as product differentiation increases. Many empirical studies test for asymmetric dynamic responses, namely the “rocket and feathers” phenomenon (Tappata 2009). Retailers may exert market power by differentiating their product portfolio to raise prices more rapidly when costs increase than they reduce prices when costs decrease. Borenstein et al. (1997, p. 324) argue that “prices are sticky downward because when input prices fall the old output price offers a natural focal point for oligopolistic sellers”. Thus, for more differentiated products, we expect retailers to adjust prices more asymmetrically.¹³

11.4.6. Price discriminations and bundling

Product differentiation and price discrimination are related, as the former is often a means to achieving the latter. Price discrimination, however, can also work without product differentiation. Price discrimination is sometimes defined as setting different prices for the same product. “However, this definition fails on two accounts: different prices charged to different consumers could simply reflect transportation costs, or similar costs of selling the good; and price discrimination could be present even when all consumers are charged the same price” (Varian 1989, p. 598). We therefore use Stigler’s (1987) definition: “a firm price discriminates when the ratio of prices is different from the ratio of marginal costs for two goods offered by a firm.”

$$\frac{P_1}{P_2} \neq \frac{\frac{\partial C}{\partial q_1}}{\frac{\partial C}{\partial q_2}} \quad (45)$$

¹³ Chapter 10 includes a detailed discussion of the causes and empirical measurement of asymmetric CPT.

Clerides (2002, 2004) points out that absolute deviations between prices and marginal costs also qualify as price discrimination, and that price discrimination can occur in the short and/or long term depending on whether one considers marginal or total costs.

$$P_1 - \frac{\partial C}{\partial q_1} \neq P_2 - \frac{\partial C}{\partial q_2} \quad \text{short term} \quad (46)$$

$$P_1 - \frac{C}{q_1} \neq P_2 - \frac{C}{q_2} \quad \text{long term} \quad (47)$$

Under perfect competition, prices should equal marginal costs in the short term and average costs in the long term. Thus, under perfect competition we do not expect price discrimination. Three conditions are necessary to successfully discriminate in terms of price. Firms need to have market power; they require knowledge of differences in consumer demand and the ability to sort customers; and they must be able to prevent resales.¹⁴

Price discrimination is defined by differences in the mark-ups of similar products. Market power is defined by the ability to enforce mark-ups. We assume that profit maximisation is the main incentive for price discrimination, but it can result from the pursuit of other objectives, such as redistribution of income or state regulation. Low-price meals for students or uniform postage charges for standard letters are examples of price discrimination that is not driven by the profit motive. University cafeterias in Germany charge students less than they charge staff and guests for the same meals, not to maximise profit but rather to indirectly support student incomes. In Germany, the cost of sending a standard letter by mail is fixed by law, regardless of the distance between sender and destination.

Figure 11.16 shows the magnitude of price discrimination in a university cafeteria in Germany. Even though the marginal cost of preparing a meal is the same regardless of who eats it, guests pay the highest prices, followed by university staff and then by students, who pay only 40% to 60% of what guests are charged. The ratio is higher for more expensive meals.

Students also get discounts at theatres, movie theatres, museums, and for public transportation. Movie theatres differentiate not only between consumer groups (student, but also children under a certain age) but also for the day of the week. The cost of admission for a regular movie in many

¹⁴ Due to self-selection, resales do not need to be prevented under second-degree price discrimination and mixed price bundling.

German movie theatres will typically be lower early in the week (Monday through Wednesday) than later (Thursday through Sunday).

Figure 11.16: Prices of meals in a German student canteen 2020.

Hauptgericht (Main course)

Vegetarian stuffed pepper with tomato sauce

Students	1,75 €
Employees	2,60 €
Guests	4,00 €

Pork cutlet with mushroom sauce

Students	1,85 €
Employees	2,70 €
Guests	4,00 €

Suppe & Eintopf (Soup & stew)

Soup of the day

Students	1,00 €
Employees	1,50 €
Guests	2,50 €

Vom Grill (Grilled)

Grilled sausage with French fries, curry catsup

Students	2,90 €
Employees	3,90 €
Guests	4,90 €

Source: <https://www.studierendenwerk-bonn.de/>.

Another example of price discrimination is provided by the book market. In March 2021, Varian’s “Microeconomic Analysis” was listed on Amazon (www.amazon.de) for €55.99 in hardcover, and for 53% less at €29.95 in paperback. Although the marginal cost of producing hardcover is higher than that of paperback, this is very likely a case of price discrimination.¹⁵

Price discrimination can affect different dimensions, e. g. in respect of customer groups, volumes purchased, qualitatively differentiated products, over time, over space, etc. We distinguish four different types of price discrimination: first, second, and third degree, and bundling (over different products).

¹⁵ According to Gorham Printing (<https://gorhamprinting.com/prices-book-printing/price-charts.html>) the cost of printing a softcover book with 300 pages including interior colour pages and a print run of 500 is \$16.20; the same book in hardcover costs \$26.08. Hence, both the absolute and the relative prices cited here for Varian’s textbook cannot be explained by marginal production costs.

First-degree price discrimination

For first-degree price discrimination, firms need to know the demand of all individual buyers and need to make sure that no reselling takes place. For each and every unit of sale, firms set a price based on the consumer's willingness to pay. By this strategy, firms skim the entire willingness to pay for each and every unit. Alternatively, firms set a take-it-or-leave-it price for a fixed quantity that equals the consumer's entire willingness to pay.¹⁶ Another strategy is a two-tier pricing scheme by which customers pay a price equal to the marginal costs plus a fixed amount (entry fee), which must be paid before purchase. First-degree price discrimination is an ideal; the required information on each individual consumer's willingness to pay and/or the ability to prevent resales are often missing. Theoretically, first-degree price discrimination by a monopolist leads to the same allocation as perfect competition; thus, there is no net welfare loss, but the monopolist receives the entire sum of producer and consumer surplus.

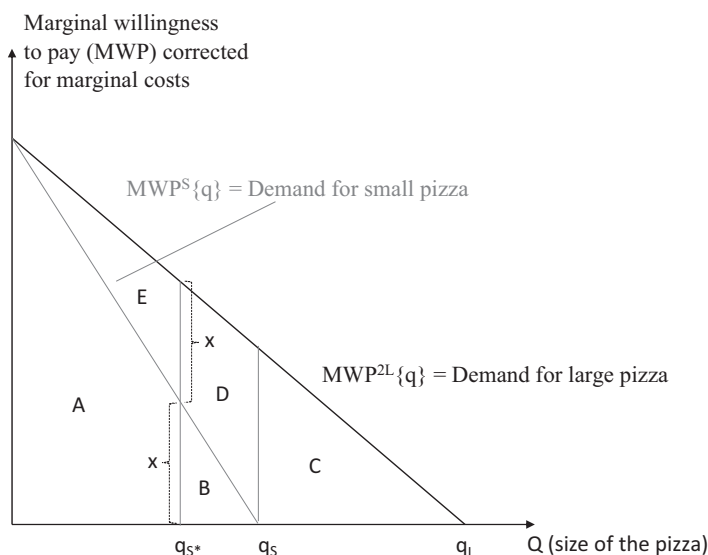
Second-degree price discrimination

"Second-degree price discrimination encompasses a variety of pricing schemes through which the firm is able to induce consumers with high valuations to pay higher prices than consumers with low valuations" (Carroll and Coates 1999, p. 469). Consumers self-select into groups based on their willingness to pay. Firms have no information on individual consumers' willingness to pay. However, they know that the willingness varies among consumers, and they have some idea of the (approximate) size of groups of consumers with different preferences, for example with respect to the quantity or the quality of the product. Examples are volume discounts or different mark-ups for different qualities, e. g. paperback and hardcover books.

Consider the following simple example of second-degree price discrimination. Assume that we run a pizza delivery service. Half of our potential customers prefer smaller pizzas and the other half larger pizzas. Demand (marginal willingness to pay) functions for these two groups are depicted in Figure 11.17. These demand functions have been corrected for the respective variable costs of production, which might vary somewhat with pizza size. Thus, the areas beneath these demand functions show the maximum profit that could be generated through sales to individual

¹⁶ Such a take-it-or-leave-it offer would already be an example for bundling over the same good or second-degree price discrimination.

Figure 11.17: Second degree (quantitative) price discrimination



Instead of only decreasing the price for large pizzas, the firm can additionally change the size of the pizzas to maximise profits. Increasing the sizes beyond q_S and q_L makes no sense as the respective willingness to

¹⁷ Under second degree price discrimination, buyers self-select into the pricing arrangement that is best for them and for the seller (Carroll and Coates 1999).

pay does not increase beyond q_S and q_L . Making large pizzas smaller than q_L has no positive effect either, because it would only reduce the area C . The only relevant option is making small pizzas smaller than q_S . As a consequence, we need to reduce the price of small pizzas, for example to A . Thus, we lose profit equivalent to area B from the small pizza lovers. However, we can now increase the price of the large pizza by $B+D$. Our net gain is $(D-B)$, which is positive. The profit $2A+B+D+C$ is maximised by reducing the size of the small pizzas until resulting marginal reduction in B equals the marginal gain in D , which is the case for q_{S^*} in Figure 11.17 (where both marginal changes equal x). This result holds as long as the shares of small and large pizza lovers are the same. If, for example, the restaurant has twice as many small pizza lovers than large pizza lovers, then the marginal loss in B needs to be half of the marginal gain in D .

Third degree price discrimination

Third-degree price discrimination is based on differences between groups of consumers in their price sensitivity. Intuitively, less sensitive consumer groups pay higher prices. Groups need to be identified and separated, and reselling needs to be restricted. Groups such as students, children, retired or handicapped individuals can easily be identified by identification cards. In some cases, identification is not necessary if groups are naturally separated by region or time (for example, customer groups who want to see a movie on different nights of the week). Assuming the supplier has information on the total demand of each group, and constant marginal costs, the problem simplifies to the monopoly equilibrium in each individual market. If marginal costs are increasing, the solution is more complicated. The necessary condition remains: marginal cost equals marginal revenue in both markets. Figure 11.18 shows an example with two groups of customers. The profit maximising quantity sold ($q^M = q^{D1+D2}$) is reached where aggregate marginal revenue ($R'\{q\}$) equals marginal cost ($C'\{q\}$). This quantity is then divided between groups 1 (q^{D1}) and 2 (q^{D2}), such that the marginal revenue of the last item sold is the same in both groups.

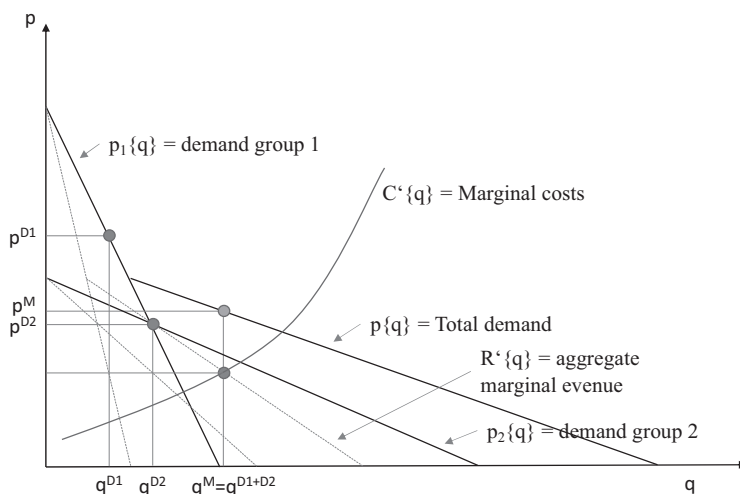
Bundling

Bundling is the practice of selling two or more goods in a package. Pure bundling is when products are only available in a package; mixed bundling is when the bundle and the individual goods are both available to the consumer. Examples are menu offers in a restaurant or in fast food chains, hardware purchases combined with service contracts, season tickets, etc. Second-degree price discrimination can be viewed as a special kind of

bundling over the same product. Bundles can be fixed or flexible. For example, some printer brands insist that consumers use same-brand toner refills. Like other forms of price discrimination, bundling is often used to maximise profits. Further arguments for bundling are efficiency in production or assembling, quality assurance, avoiding price regulations, promotion of (new) products, and price-fixing agreements.

Sometimes it is more efficient to sell bundles. There are strong economies of scale associated with the production of many manufactured goods such as automobiles, kitchen appliances and cameras; few, if any, consumers would consider assembling these products themselves, even though the component parts could, at least in principle, be purchased separately. Many such products can thus be viewed as bundles. Some firms sell complements in bundles or require the follow-up purchase of another product to activate a quality agreement or service guarantee. Examples are car brands that require certain operating supplies or analogue cameras which required film rolls and processing services. Kodak, for example, gave away free tickets for processing with each film roll sold to ensure a certain quality standard. Another reason for bundling is to avoid price regulations. If a firm is limited by a minimum price regulation or cartel agreement, selling product bundles enables it to mask individual prices and thereby avoid the price regulation and possible associated penalties.

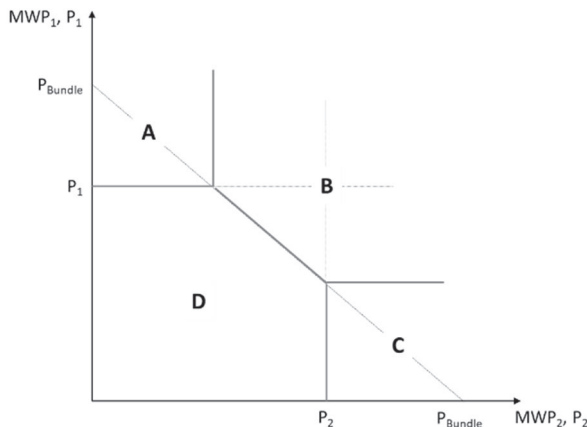
Figure 11.18: Third degree price discrimination, monopoly, and perfect competition.



If the supplier is a profit maximiser, bundling works well if the consumers' willingness to pay is negatively related. If one consumer is willing to pay more for one and less for the other product than another consumer, then bundling is the best way to extract the consumers' willingness to pay. Assume, for example, that consumer 1 is willing to pay €12 for good 1 and €8 for good 2, and consumer 2 is willing to pay €15 for good 1 and €5 for good 2. A monopolist selling these products separately would charge €12 for good 1 and €5 for good 2. If the two consumers are prepared to buy at most one unit of each product and assuming zero production cost, the profit is €34. However, if the monopolist bundles and sells the goods together for €20, both consumers will buy the pure bundle and the total profit is €40.

Mixed bundling is more complicated due to the additional alternative of buying the single products. Figure 11.19 is a schematic depiction of the situation. The monopoly firm chooses the individual prices (P_1 and P_2 for goods 1 and 2, respectively) and the bundle price (P_{Bundle}) in order to maximise profits. The consumers choose depending on their willingness to pay (WTP). Area *A* in Figure 11.19 contains the WTP combinations for which consumers would only buy product 1. Area *C* is the corresponding area for good 2. Area *D* contains the WTP combinations for which consumers do not buy at all, and area *B* those for which consumers choose to buy the bundle. The monopoly firm can change the relative sizes of these areas by adjusting the prices for the single products, and by moving P_{Bundle} closer to or farther away from the origin.

Figure 11.19: Mixed bundling



Source: Own based on Carlton and Perloff (2015:350).

The following example illustrates an optimisation problem with two goods and three consumers. The monopoly supplier has constant marginal costs of production of €6 per unit of good 1 and €10 per unit of good 2. The willingness to pay for the two goods (good 1/good 2) is €20/€4 for consumer 1, €16/€8 for consumer 2, and €8/€16 for consumer 3. If we apply fixed bundling, the best bundle price is €24. All three consumers would buy the bundle and the profit is: $(3 \times €24) - (3 \times €10) - (3 \times €6) = €24$. If the monopolist sells the products separately, the optimal prices are €16/€8 resulting in a profit of: $2 \times (€16 + €8) - (2 \times €10) - (2 \times €6) = €16$. The optimal mixed bundling strategy would be to sell consumer 1 only good 1, to sell consumer 2 the bundle, and to sell consumer 3 only good 2, with the bundle price set at €24, the price of good 1 at €19.99, and the price for good 2 at €15.99. The profit under mixed bundling is then: $€19.99 + €24 + €15.99 - (2 \times €10) - (2 \times €6) = €27.98$.

11.4.7. Information and models of sales

We live in an information society. It has never been easier to obtain information about all manner of things. Often not a lack of information but rather information overload causes difficulties in decision making, for example in consumer choice. Everybody individually relates to the term information, but what exactly is it? According to Shapiro und Varian (1998, p. 3) information is “anything that can be digitised - encoded by a stream of bits”. Thus, many things may be viewed as information. Information often has a positive connotation, as something that is objective, correct, or useful. However, this is not necessarily the case. Key properties of information include: (1) not all information is correct, (2) not all information is processed, (3) information gathering and processing is costly, (4) not all information is readily available at all times, (5) information can change over time, (6) not everybody can correctly process each piece of information, (7) information is subjective, and (8) information may not be structured and may even be without value (Case 2012, p. 45 ff.).

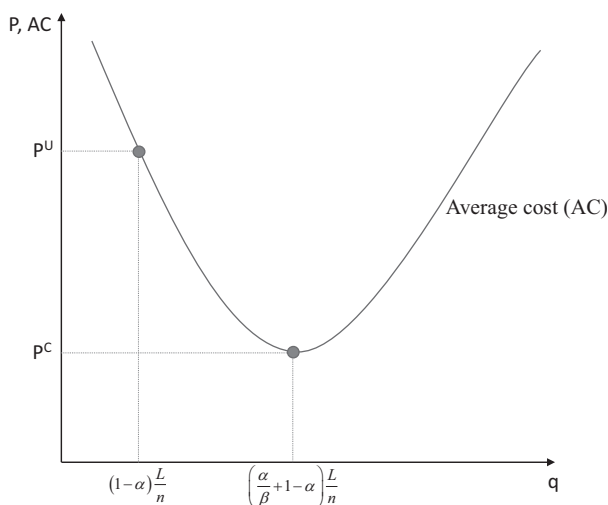
Information on prices, product quality, quantities, etc. is relevant for market participants. In his paper “The Market for Lemons”, for example, Akerlof (1970) analyses the implications of imperfect (asymmetric) information about product quality. He shows that there will be no market for high quality goods unless, for example, sellers offer product guarantees or warranties. In the following, we focus on the role of price information. To simplify the issue, we first assume that information is just about the product price, it is always correct, and all agents can correctly process it. Time is not an issue.

However, information gathering is costly and/or consumers have different levels of price information (knowledge). The motivation of the following models is to demonstrate the possible existence of multi-price equilibria under this setting. Salop and Stiglitz (1977) first present a model of the role of information, which results in a two-price equilibrium with high- and low-price stores. Varian's (1980) model of sales produces mixed pricing strategies, with a continuum of prices in the range between low promotional sales' prices and high reservation prices. We begin with Salop and Stiglitz's (1977) model of bargains and rip-offs.

11.4.7.1. Bargains and rip-offs (Salop and Stiglitz 1977):

Salop and Stiglitz (1977) analyse the effect of information costs in the context of (food) retailing. They assume two types of consumers. One type has low marginal costs of gathering and processing price information (c_1), and the other has high costs ($c_2 > c_1$). The model assumes one identical product. Each consumer demands a maximum of one unit of this product, and each is willing to pay at most P^u for this unit. There are n firms, manufacturers or retailers who have free entry into the market. Thus, profits are zero. The average production and/or retailing cost curve (AC) is U-shaped and identical for all firms (Figure 11.20).

Figure 11.20: The Salop and Stiglitz (1977) model of bargains and rip-offs



Source: Own depiction based on Salop and Stiglitz (1977).

The minimum of the average cost curve is at P^C . If consumers invest the costs of gathering and processing information, they become informed about all prices in the market. If not, they do not know any prices charged by firms in the market. Though the location of the firms is not relevant, each consumer can only buy from one firm. Informed consumers buy from the firm with the lowest price. Consumers with no price information randomly choose a firm. The share of informed consumers is α and the share of uninformed consumers is therefore $1-\alpha$. The number of consumers in the market is L .

Based on the model assumptions, three different price equilibria can result. First, when all consumers are informed, the price P^C results. To prove this, we start at a price above P^C . If all firms set a price above P^C , each firm has an incentive to marginally lower the price to attract all informed consumers. Because the price decrease is only marginal and average costs fall, demand and profits increase significantly. Thus, firms will underbid each other down to P^C . No firm will bid below P^C , which represents the minimum average cost.

The second price equilibrium is P^U , the maximum WTP common to all consumers. A price above P^U would lead to zero demand, zero sales, and zero profit. If all consumers are uninformed due to high costs of information, consumers randomly choose firms. In this situation, a price decrease would not generate an increase in demand for a firm because there are no informed consumers. Therefore, the profit-maximising price is at P^U .

Third, a two-price equilibrium can occur if some consumers are informed and others are not. This equilibrium occurs if the following condition is met:

$$c_1 < (1 - \beta)(P^U - P^C) < c_2 \quad (48)$$

Consumers search because the reward of the search exceeds its costs. In this case, the reward takes the form of finding the low-price firm. A potential equilibrium with two prices in the market is represented by a low price at P^C and a high price at P^U . Prices above P^U and below P^C are not set by firms because there is no demand above P^U and costs exceed prices below P^C . Prices between P^U and P^C are suboptimal because firms that set such prices would not attract informed consumers, and therefore, they would be better off charging P^U . Only firms that charge P^C share the informed consumers. Thus, if a two-price equilibrium exists, then because some firms set P^U and others P^C . Due to free entry, profits are zero for all firms. Average costs at the demand of high-price firms need to equal P^U and average costs at the

demand of low-price firms need to equal P^C . The demand (the number of consumers expected) at the high-price firms is equal to the share of uninformed consumers times the total number of consumers divided by the number of firms:

$$d_{pU} = (1 - \alpha) \frac{L}{n} \quad (49)$$

The demand at the low-price firms is the demand at the high-price firm plus an even share of the informed consumers at low-price firms, where β is the share of low-price firms.

$$d_{pC} = (1 - \alpha) \frac{L}{n} + \alpha \frac{L}{n\beta} = \frac{((1-\alpha)\beta + \alpha)L}{n\beta} \quad (50)$$

By substituting the demand at high-priced firms in equation (49) into equation (50), we can determine the share of low-price firms β :

$$\beta = \frac{\alpha d_{pU}}{(1-\alpha)(d_{pC} - d_{pU})} \quad (51)$$

The share of low-price firms increases with the share of informed consumers and with the individual demand at high-price firms; in addition, the share of low-price firms decreases with the share of informed consumers and the demand at low-price firms.

Salop and Stiglitz show that it is possible to have a two-price equilibrium for a homogeneous good market. The retail food market, where we observe significant product price differences between low-priced discounters and high-priced supermarkets, is a good example. However, the model cannot explain price differences between products within a retail store, nor can it explain price changes over time, e. g. promotional prices. In addition, the model does not explain which store sets what price and why, because the decision-making process at the store level is not modelled. Finally, the model is dynamically inconsistent. Uninformed consumers randomly visit stores. Thus, some of them will find a low-price store by chance. After finding such a store, consumers will return to it in the future. Thus, in the long run, all consumers will become informed either systematically or randomly, and the two-price equilibrium will dynamically converge to a one-price equilibrium at P^C with only informed consumers. Varian (1980) takes up this critique in the model that we consider next.

11.4.7.2. *A model of sales (Varian 1980):*

In the late 1970s Hal Varian shopped for a new TV and noticed that there were substantial price differences and special offers. Inspired by these observations, Varian developed a model of (promotional price) sales. Varian (1980, p. 651) states “Economists have belatedly come to recognise that the “law of one price” is no law at all. Most retail markets are instead characterised by a rather large degree of price dispersion”.

Varian (1980) and Salop and Stiglitz (1977) make very similar assumptions. The main difference is the assumption of a monotonically decreasing average cost curve. Salop and Stiglitz (1977) assume a U-shaped average cost curve with a low-cost minimum. Varian argues that retailing costs are, for the most part, fixed and that as a result, average retailing costs monotonically decline with sales volume. Instead of introducing search costs, Varian assumes a fixed share of informed and uninformed consumers.

Following Varian’s (1980) notation there are n firms or retail stores, and market entry is free. There are I informed and M uninformed consumers. Each consumer demands at most one unit of the homogeneous good. Uninformed consumers randomly choose a store; thus, every store gets the same share ($U = M/n$) of uninformed consumers. The maximum WTP, or reservation price for the good (r), is the same for each consumer. All informed consumers choose the lowest-price store. The number of consumers at the store with the lowest price is $U + I$.

For these assumptions, Varian (1980) shows that there exists no symmetric price equilibrium (not all stores set the same price). Instead, an equilibrium can only exist for a mixed pricing strategy based on a continuous price distribution (probability density function, $f(P)$). Figure 11.21 shows the setting for the Varian (1980) model. First, we define the maximum price range. The maximum price is equal to the reservation price (r). The minimum price equals the average costs for a sales volume of $I + U$ units and is marked P^* in Figure 11.21. Neither r nor P^* represent symmetric price equilibria. If all stores offer at price r , then every store has an incentive to underbid competitors in order to attract the informed shoppers. If all firms set P^* , then all stores make losses as they will each attract an equal share of the informed consumers ($U + I/n$). Varian (1980) argues that there can be no mass (price) points in the interval between r and P^* . If a mass point in the distribution of prices did exist, stores would have an incentive to underbid this price in order to attract all informed consumers. Therefore, the price distribution must be continuous.

To derive the probability density function of P ($f(p)$), we follow Varian's (1980) initial idea and derive the expected profit ($\pi(P)$) of a store that sets price P :

$$E[\pi(P)] = \pi_s(1 - F(P))^{n-1} + \pi_f(1 - (1 - F(P))^{n-1}) \quad (52)$$

$F(P)$ is the cumulative probability density function or the integral of $f(P)$. Figure 11.21 illustrates the expected profit if the firm sets price P . In that case, we have two potential outcomes. First, if the store setting price P is the lowest price store, it gets $I + U$ consumers and the profit amounts to the area $\pi_s = (P - P^*)(U + I) > 0$. If the store's price P is not the lowest price in the market, then the store only sells U units at the average cost of r and makes the profit $\pi_f = (P - r)U < 0$; thus, the retailer makes a loss equivalent to area π_f . The expected profit is the sum of the π_s times the probability that P is the lowest price in the market ($(1 - F(P))^{n-1}$) and π_f times the residual probability $(1 - (1 - F(P))^{n-1})$. The probabilities depend on P . If the firm sets a higher price, the probability that it will be the low-price supplier decreases. To calculate the probability of being the lowest price store, we start with the probability that a competitor sets a price above P . The probability that one other store sets a price above P is $1 - F(P)$. The probability that all other stores set prices above P is $(1 - F(P))^{n-1}$, which is the probability that the store setting price P is the lowest price store. The probability that the store is not the lowest price store is $(1 - (1 - F(P))^{n-1})$. Because of free entry, the expected profits of all firms at all prices in the range between r and P^* are zero.

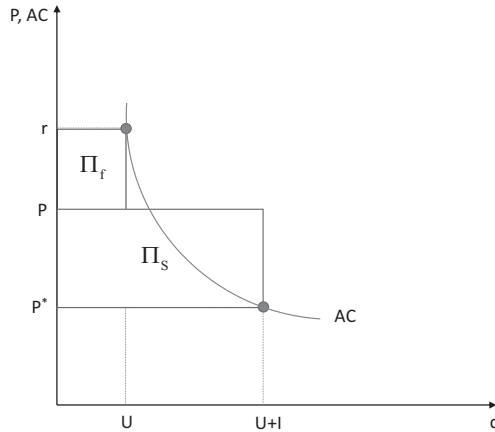
If we set the expected profit to zero, we can rearrange equation (52) which results in:

$$F(P) = 1 - \left(\frac{\pi_f}{\pi_f - \pi_s} \right)^{\frac{1}{n-1}} \quad (53)$$

The store's profit at price P is negative when P is not the lowest price in the market. The store's profit at price P is positive, when P is the lowest price. Thus, the term in brackets in equation (53) is positive and smaller than 1. The likelihood that P is the lowest price depends on the ratio of the firm's profits at P , when P is the lowest and when P is not the lowest price in the market. For higher prices, π_f is decreasing and π_s is increasing; thus, the ratio tends to zero. For lower prices, π_f increases and π_s decreases and the

ratio approaches 1. From equation (53) we can derive the equilibrium price probability density function.

Figure 11.21: The Varian (1980) model of sales



Source: Own depiction based on Varian (1980, p.655).

For $P = r$ we can conclude that the price cannot be the lowest in the market and thus the profit is equal to $r \cdot M/n - k$, where k are the fixed costs. Due to free entry, the profit even in this extreme case is zero and therefore $U = k/r$. The other extreme is $P=P^*$, which is the lowest price in the market. In this case the profit is $P^*(I + k/r) - k = 0$; thus $P^* = k/(I + k/r)$. We introduce the general profit equations $((PU - k), P(U + I) - k)$ to equation (53) and simplify:

$$F(P) = 1 - \left(\frac{PU - k}{PU - k - P(U + I) + k} \right)^{\frac{1}{n-1}} \quad (54)$$

$$F(P) = 1 - \left(\frac{PU - k}{-PI} \right)^{\frac{1}{n-1}} \quad (55)$$

$$F(P) = 1 - \left(\frac{k - \frac{k}{r}P}{PI} \right)^{\frac{1}{n-1}} \quad (56)$$

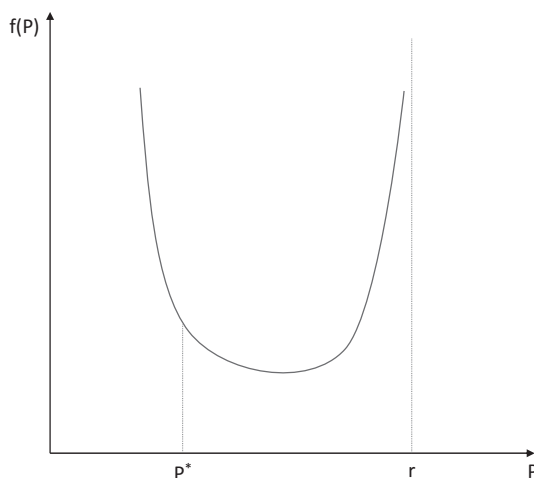
$$F(P) = 1 - \left(\frac{k}{I} \left(\frac{1}{P} - \frac{1}{r} \right) \right)^{\frac{1}{n-1}} \quad (57)$$

Next, we differentiate equation (57) with respect to P . After some rearrangements, and assuming that n goes to infinity, Varian (1980) shows that the price density is proportional to the expression on the right-hand side of the following equation:¹⁸

$$F'(P) \equiv f(P) \propto \frac{1}{P(1-\frac{P}{r})} \quad (58)$$

The density in equation (58) is U-shaped as shown in Figure 11.22, indicating high probabilities for low as well as for high prices.

Figure 11.22: Equilibrium price density for the Varian model



Source: Own depiction based on Varian (1980, p. 656).

The Varian (1980) model can explain price differentiation at the same store in time. The model produces high and low prices as an outcome of a mixed strategy equilibrium. However, it is unrealistic that retail managers would set prices by randomly drawing from a distribution every day or week. Further, the focus on one product oversimplifies the market situation in retailing, where there are thousands of products per store, as well as the interactions between them in demand and supply. Richards (2006) develops a multiproduct model of sales based on Varian (1980) to prove the existence

¹⁸ A detailed derivation of Varian's (1980) results is presented in the Appendix to this chapter.

of a complementary relationship between breadth and depth of sales. Loy and Weiss (2009), however, show that the model results in a substitutional relationship; thus, the number of sales and the average level of discounts are substitutes.

In the following, we add a second product to study multiple products and the role of brands and brand loyalty in price setting and promotion.

11.4.8. The effects of brand loyalty (Raju et al. 1990):

At the beginning of this chapter we saw that brands play a major role in food retailing. By establishing brands, manufacturers and retailers try to differentiate the food retail market horizontally and/or vertically. “A brand is a name, term, design, symbol or any other feature that identifies one seller's good or service as distinct from those of other sellers” (Muñiz 2015). A brand “is an intangible asset” that is intended to create “distinctive images and associations in the minds of stakeholders, thereby generating economic benefit (value)”. In practice, strong brands attract customers, and add revenue through increased price or volume premiums and repeat purchase loyalty. Brands can create economic benefit, if a brand producer can raise prices above competitors’ level without losing too much sales volume. Brand loyalty is the key term behind the strength of brands. According to Jacoby and Kyner (1973, p. 2) “brand loyalty is a biased (non-random) behavioural response expressed over time by some decision-making unit with respect to one or more alternative brands out of a set of such brands and is a function of psychological (decision making, evaluative) processes.” Brands add extra value to the product, for which the consumer is willing to accept a higher price. Agrawal (1996, p. 87) defines the “strength of brand loyalty ... as the price differential needed before the consumer will switch away to the less favoured brand”. Hence, a consumer is loyal with an intensity l_i to brand i if he/she does not switch until:

$$P_{i \neq 1} < P_1 - l_1 \text{ with } l_1 > 0 \quad (59)$$

Raju et al. (1990) built on this definition and developed a model of brand loyalty with two competing products. They assume two producers, one of which markets a strong (s) brand, while the other markets a weak (w) brand, with $l_s > l_w$. There are two groups of customers, one of which is loyal to the weak brand while the other is loyal to the strong brand. The reservation price is r for all consumers, and every consumer demands at most one unit of each good. Raju et al. (1990) abstract from production and transaction

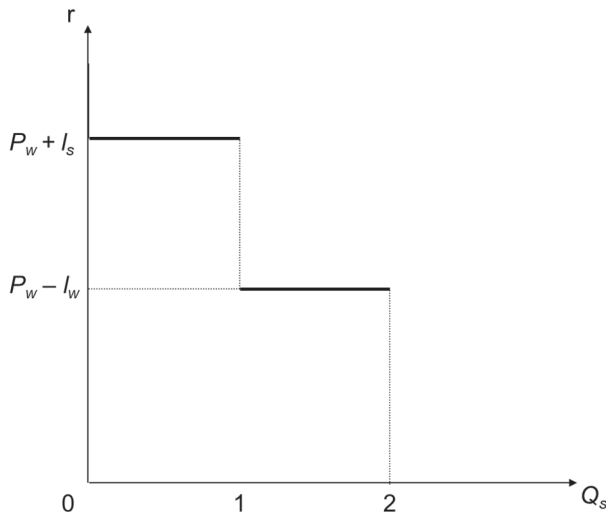
costs. The demand functions for the weak and for the strong brands are shown in equations (60) and (61), respectively.

$$q_s(p_s, p_w) = \begin{cases} 0 & \text{if } p_w < p_s - l_s \\ 1 & \text{if } p_w - l_w \leq p_s \leq p_w + l_s \\ 2 & \text{if } p_s < p_w - l_w \end{cases} \quad (60)$$

$$q_w(p_s, p_w) = \begin{cases} 2 & \text{if } p_w < p_s - l_s \\ 1 & \text{if } p_w - l_w \leq p_s \leq p_w + l_s \\ 0 & \text{if } p_s < p_w - l_w \end{cases} \quad (61)$$

Figure 11.23 illustrates the demand for the strong brand. Like Varian (1980), Raju et al. (1990) find a mixed pricing strategy for both firms. The main result is that the strong brand promotes less often and at higher discounts than the weak brand. Promotion of the strong brand is used to attract consumers loyal to the weak brand (aggressive promotion), while promotion of the weak brand is used to discourage its loyal customers from switching to the strong brand (defensive promotion). Figure 11.24 shows the equilibrium probability density functions for both brands.

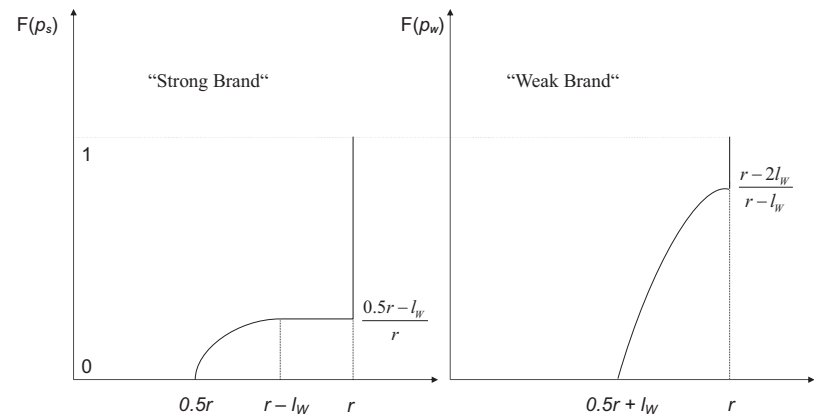
Figure 11.23: Demand for the strong brand in the Raju et al. (1990) model



Source: Own depiction based on Raju et al. (1990).

Agrawal (1996) extends this model by introducing a retailer as an intermediary between the producers and the consumers. Agrawal's (1996) model generates significantly different outcomes, for the following reason. In the Raju et al.'s (1990) model there is no coordination of promotion between manufacturers. In Agrawal (1996), however, the monopolistic retailer has full control over promotional sales. For the retailer it is not profitable to defend customers loyal to the weak brand against promotions of the strong brand. The retailer only sets price promotions in order to encourage consumers to switch to the promoted brand. The retailer sets price promotions for both brands, but not at the same time; the discount for the strong brand is lower than the discount for the weak brand, according to the intensity of consumer loyalty. Compared with the weak brand, the strong brand needs less (price) incentive to attract the consumer of the competing (weak) brand. The producer of the strong brand will offer a better deal for discounts than the producer of the weak brand. Thus, the retailer will promote the strong brand more often than the weak brand, but at lower discounts. This result is exactly opposite to that of Raju et al. (1990). The five series of orange juice prices in the same retail store presented in Figure 11.25 appear to confirm one of Agrawal's (1996) results, as they show no evidence of synchronised brand promotions.

Figure 11.24: Probability density functions for the Raju et al. (1990) model



Source: Own depiction based on Raju et al. (1990, p. 287).

In this section we have introduced some models that aim to explain why price promotions are used in food retailing. We have seen that these models shed some first light on the complex problem of explaining the phenomena

in retail price setting. In the next section we present a number of specific case studies of price behaviour in food retailing to further explore empirical evidence in this field.

11.5. Case studies

11.5.1. Why don't prices rise during periods of peak demand (Chevalier et al. 2003)?

Chevalier et al. (2003) examine retail and wholesale prices for a large US supermarket chain over a period of more than seven years. They find that prices on average fall during seasonal demand peaks for a product, such as holidays, largely due to changes in retail margins. Retail margins for specific goods fall during peak demand periods, even if these periods do not coincide with aggregate demand peaks for the retailer.

Chevalier et al. (2003) present different theories to explain this countercyclical pricing: a model of countercyclical collusion, a model of cyclical elasticity of demand, and a model of loss leader. All of these models are potentially consistent with seasonally falling food retail prices for particular products at demand peaks.

Economies of scale in search

Warner and Barsky (1995) model economies of scale in consumer search. The costs of search and travel between retail stores are fixed for consumers. Thus, when demand is high, fixed costs are spread over a higher purchase volume and consumers search more. This makes consumers more price sensitive, leading to lower retail margins and prices. Therefore, prices tend to fall for example at Christmas or Easter, when overall demand is at a peak.

Countercyclical collusion model

Another explanation is based on dynamic interactions between firms. If retailers compete repeatedly, tacit collusion can emerge. Rotemberg and Saloner (1986) consider tacit collusion under variable demand. In periods of high demand, the temptation to defect and increase market shares by lowering prices is highest. Moreover, the possible resulting punishment for defecting is lowest, because it will occur in later periods of low demand. Bernheim and Whinston (1990) extend this analysis to the multiproduct environment and prove that prices of various products fall during periods of peak demand.

Loss-Leader model

The Lal and Matutes (1994) loss-leader model (see section 3.3) provides another explanation of falling prices in periods of peak demand. The model assumes costs of shopping and costs of advertising. Furthermore, it assumes that consumers do not have any information on retail store prices, unless they are advertised. To attract consumers, retailers advertise low prices for some products and maintain regular high prices for the others. Consumers choose a store based on advertised prices. Because of advertisement costs, retailers do not advertise all prices, but rather only those with a high demand. Thus, in periods of high demand more products are chosen for advertisements than in periods of normal demand.

Chevalier et al.'s (2003) empirical results show that prices, and retail margins, fall in periods of peak demand, but only for seasonal products. Estimated price elasticities for these products also display significant seasonal variation. Furthermore, the authors find that advertising activity and seasonal demand are positively correlated, i. e. seasonal products are more advertised in peak demand periods. These empirical results are consistent with Lal and Matutes' (1994) loss-leader theory of retailer competition. Models that stress cyclical demand elasticities or cyclical firm conduct are less consistent with the empirical findings. Hence, it appears that manufacturer behaviour plays a limited role in the countercyclicality of prices.

11.5.2. Sales: Tests of theories on causality and timing (Berck et al., 2008)

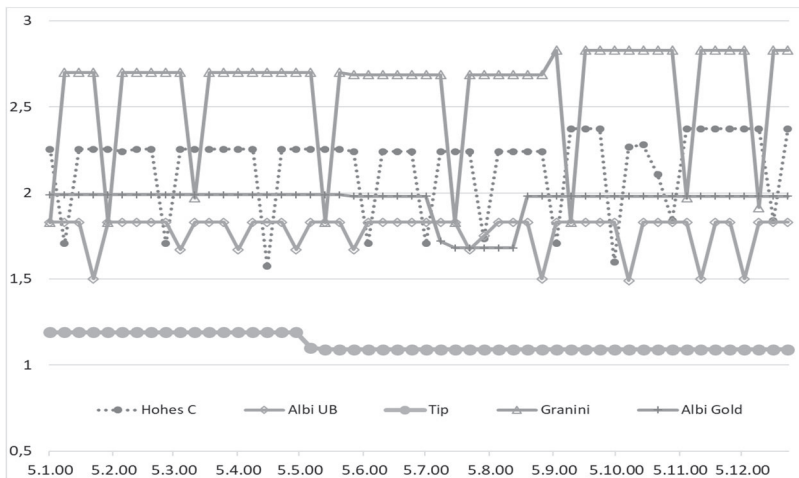
Modern theories of promotional sales make controversial predictions about the temporal pattern of sales. Berck et al. (2008) analyse grocery scanner data for orange juice in the USA to test these theories. Using correlations, runs tests, probit regressions, and vector-autoregressive analyses, they investigate the following hypotheses: (1) private label products are less often on sale than national brands; (2) manufacturers determine sales rather than retailers; (3) price distributions over longer time periods are (a) continuous (U-shaped, bell-shaped), (b) have a mass point at the highest price, or (c) have two mass points for each major brand and a single mass point for the private label; (4) price distributions vary with the perishability of the product; (5) regarding the timing of sales across firms: (a) sales patterns are random across firms, (b) firms lower prices simultaneously, (c) national-brand firms alternate sales, but private labels do not have sales, (d)

the probability of a sale rises with the time since the last sale of the relevant brands, (e) the largest brand is a pricing leader, (f) prices fall slowly and then jump to a high level.” (Berck et al. 2008, p. 1260).

The authors use retail prices of major brands and private labels of orange juice in 22 stores in 13 US cities collected by IRI Symphony for 1998 and 1999. The authors include a wholesale price for orange juice and a consumer price index in their analysis. The results show that retail prices vary strongly across stores and over time. The variation in both dimensions is largely due to temporary price promotions for individual brands. The authors define promotional sales in a store as price reductions of at least 25%, 35% or 50% against the mode price of the product in the store calculated for the two-year period.

Berck et al. (2008) find that private label orange juice is put on sale at least as frequently as the manufacturer brands. Thus, they reject the first hypothesis. However, Loy and Schaper (2014) find the opposite result for a range of products in German grocery stores: retailer brands are put on sale significantly less frequently than national brands. Figure 11.25 shows an example for 5 different orange juice brands in a German retail store in 2000. While we observe promotional price sales for three of four manufacturer brands, we observe none for the private label TIP.

Figure 11.25: Weekly retail prices of orange juice in a German food store



Source: Own depiction with price data from MaDaKom 2002.

Berck et al. (2008) also investigate the correlations of promotions for the same brands in different stores at the national, market and chain levels. They find relatively low level of correlations. For refrigerated juice, the level of correlations increases from national to chain level, but for frozen juice this is not the case. Berck et al. (2008) therefore conclude that manufacturers do not determine the timing of sales for refrigerated or frozen orange juice.

For the distribution of prices Berck et al. (2008) reject hypotheses (a) and (b). They also find little evidence for hypothesis (c) that the distribution is bimodal with one mass point at the sale's price and another mass point at the regular price. Berck et al. (2008, p. 1265) conclude that "price distributions differ in no obvious pattern by brand and by store. Little support was found for any one of the three clearly predicted price distributions." In the case presented in Figure 11.25, three of four manufacturer brands show a clear bimodal distribution with the sales' price and the regular price as mass points. The private label shows a unimodal distribution at the regular price.

For hypothesis 5 (f) Berck et al. (2008) show that prices do not fall slowly before they rise to the highest price. For the prices in Figure 11.25 it is clear that prices rise and fall equally fast. Similar to Villas-Boas (1995), Slade (1994), and Pesendorfer (2002), Berck et al. reject the null hypothesis 5 (a) that promotional price sales are random. They find some evidence that sales are intentionally dispersed between brands. Figure 11.25 supports this finding, as we cannot observe many overlaps between brands' sales. In most cases, there is only one brand on sale at any given time and all sales last only one week. Similarly, Berck et al. (2008) fail to confirm hypothesis 5 (d) that the probability of a sale increases with the time elapsed since the last sale. In addition, the price leader hypothesis 5 (e) cannot be confirmed.

11.5.3. Brand loyalty (Empen et al., 2015)

Following Narasimhan's (1988) seminal work, many research papers have modelled the pricing strategies used by retailers to exploit brand loyal consumers. Empen et al. (2015) merge consumer and retail scanner data sets to provide a comprehensive basis for investigating the relationship between measures of brand loyalty and retailers' price promotion strategies. The study contributes to previous work by analysing two different dimensions of brand loyalty (size and strength) for the ready-to-eat cereal market.

Theoretical as well as empirical models employ a reduced brand architecture (product line management) that is focused on one brand for

each manufacturer (corporate brand). The models arrive at different conclusions depending on the various assumptions. For example, Koças and Bohlmann (2008) find that the stronger brands are promoted more aggressively, while, as discussed in section 11.4.8, Agrawal (1996) predicts that they will be promoted less deeply but more often. In the real world, manufacturers often establish a corporate brand with several sub-brands, something that is not addressed in the existing models. In the sample analysed by Empen et al. (2015), each corporate brand sells several sub-brands. For example, Kellogg's sells Frosties and Special K; Nestlé sells Trio and Fruit Loops. This type of product line management can be found in other product markets such as soft drinks (consider for example the range of Coke products), yoghurts or automobiles. Empen et al. (2015) distinguish between corporate and sub-brands by introducing corporate brand dummies. They find that Kellogg's sub-brands on average capture a larger loyal consumer segment and a higher strength of loyalty than Nestlé's sub-brands. According to the theory of Koças and Bohlmann (2008), Nestlé follows an aggressive promotional strategy by offering more sales at a higher discount because Nestlé sells to more customers who switch (higher switching to loyal ratio). Hence, Nestlé relies on being the cheapest alternative and may therefore offer more aggressive trade deals to retailers.

Empen et al. (2015) capture corporate brand effects with dummy variables; the remaining variation in promotional strategies indicates the variations between the different sub-brands. While some sub-brands, e. g. Kellogg's Frosties, have a large loyal consumer segment, the strength of the loyalty is comparatively low. Other sub-brands, e. g. Kellogg's Toppas Classic, have a smaller but strongly loyal consumer segment. These differences between sub-brands determine the relationship between brand loyalty and price promotions. Empen et al. (2015) find strong evidence that the loyal segment size has a positive impact on promotional measures, whereas the strength of consumer loyalty negatively impacts the depth and the frequency of sales. As can be seen in Table 11.5, the segment size positively affects the depth of discounts (0.062) and the frequency of discounts (0.263). The loyalty strength has a negative impact on both measures, -0.075 and -0.283 respectively.

To date, there is no consistent theory in the literature that explains this result. Empen et al. (2015) hypothesise that it is caused by the retailers' strategy to choose sub-brands for promotional sales. If retailers use brand promotions mainly to lure customers into their stores, then they will choose sub-brands that have large loyal segments; this ensures that the promotion will attract a large number of customers. However, retailers will not choose sub-brands

with strong brand loyalty, because they would rather exploit strongly loyal customers through high prices. Promotions for such sub-brands are costly, as prices have to be reduced heavily to reach new customers. Thus, there are many promotions with higher discounts for sub-brands with large loyal segments and only few promotions with low discounts for sub-brands with strong loyal segments. Price promotions for these sub-brands are not motivated by loss leadership but are rather attempts to increase the sub-brands' loyal segments to prevent customers from switching to competitors' brands.

The results suggest that weaker corporate brands, such as Nestlé, are promoted more aggressively, which supports the model hypotheses by Koças and Bohlmann (2008) (see the estimates for corporate brand dummies in Table 11.5). Within the manufacturer's product line, sub-brands with larger loyal consumer segments and moderate consumer loyalty are more often used for effective price promotions, which reflects the idea of loss leading.

Table 11.5: Results for the frequency and depth of price promotion models

Independent variables	Dependent variable	
	Depth of discount sales (Tobit model)	Frequency of discounts (OLS estimation)
Constant	0.116 (7.89)	-2.924 (-20.39)
<i>Brand loyalty</i>		
Loyalty strength	-0.075 (-3.71)	-0.283 (-1.74)
Segment size / 1000	0.062 (7.98)	0.263 (3.48)
<i>Retailer dummies</i>		
Markant	-0.022 (-2.28)	-0.156 (-1.93)
Metro	0.001 (0.15)	0.161 (2.38)
Rewe	-0.173 (-13.03)	-0.139 (-1.09)
Tengelmann	-0.039 (-2.67)	-0.792 (-8.57)
Discounter	-0.173 (-8.07)	-0.902 (-6.23)
Others	-0.016 (-0.98)	-0.869 (-7.98)
<i>Corporate brand dummies</i>		
Kelloggs	-0.025 (-2.75)	-0.212 (-1.77)
Koelln	-0.091 (-4.46)	-0.913 (-6.34)
Nestlé	0.039 (4.17)	0.503 (4.37)
Number of observations	1729	1012
R ²		0.23

Note: *t*-values in brackets.

Source: Empen et al. (2015, p. 747).

11.5.4. Product differentiation and cost pass-through (Bittmann et al. 2020)

Many studies that estimate CPT rates or vertical price transmission for agricultural and food product markets focus on short-run dynamics, and in particular on asymmetric ‘rocket and feathers’ adjustments of prices to shocks.¹⁹ Loy and Weiss (2019) introduce a measure of product differentiation into a reduced form model of retail CPT. The analysis in the paper by Bittmann et al. (2020) builds on Loy and Weiss (2019), and uses an oligopoly model approach to derive and empirically estimate the effects of product differentiation on CPT for industry-wide and firm-specific cost shocks. Theory points to the importance of distinguishing between industry-wide and firm-specific cost shocks and their interactions with product differentiation in the context of retailers’ price-setting behaviour (see section 4.5 in this chapter).

Bittmann et al. (2020) analyse CPT in differentiated oligopolies using data on ready-to-eat soup products in the Canadian retail market that includes detailed information on product-specific purchase prices. This allows them to differentiate between and estimate industry-wide and firm-specific costs. The availability of detailed UPC-level product and brand information allows them to construct and estimate several distance measures of product differentiation, and to merge these variables with a large panel of purchase and retail prices. Using these data, they test the following hypotheses: (1) more differentiated products show higher retail prices, (2) more differentiated products have a lower CPT elasticity of industry-wide cost shocks, (3) more differentiated products have a higher CPT elasticity of firm-specific cost shocks, (4) more differentiated products return more slowly to the long-run price-cost equilibrium, and (5) product differentiation reinforces the asymmetric CPT and slows down the price adjustment process.

The results in Table 11.6 provide support for hypotheses 1 to 5. Product differentiation is associated with higher margins; CPT rates for industry-wide cost shocks are lower for more differentiated products; CPT rates for firm-specific cost shocks are higher for more differentiated products; retail prices adjust less rapidly to cost shocks for more differentiated products; and for more differentiated products the ‘rocket and feathers’ phenomenon becomes more pronounced. These results confirm that product

¹⁹ A detailed discussion of asymmetric vertical price adjustment is provided in Section 6 of Chapter 10.

differentiation serves as a tool of market segmentation to reduce price competition, and that it is a source of retail market power which directly effects product-specific mark-ups and CPT rates.

11.6. Outlook

This chapter presents an overview of theories and facts about food retail pricing. Some aspects have been covered briefly, and others in more detail. However, many aspects have been left out due to space limitations. In this final overview, some relevant missing aspects are mentioned.

First, promotional sales are key measures of food retail marketing. Many scientific papers analyse the effects of promotional prices on sale volumes, consumer loyalty, stockholding, brand switching etc. Gedenk (2002) and Blattberg and Neslin (1990) provide excellent introductions to the literature in this field.

Second, the literature is dominated by studies that use data from US food retail markets. More studies need to be carried out on markets in other countries, and the results need to be compared to deepen our understanding of food retail pricing. Among the few such comparative studies, Lamey et al. (2007) compare private labels in the US and Europe.

Table 11.6: Fixed and random effects panel model of long- and short-run CPT

Long-run CPT		Random Effects Model	Short-run asym. CPT		Random Effects Model
Dependent variable: retail prices			Dependent variable: First differences of retail prices		
Independent variables			Independent variables		
Measure of Product differentiation (PD)		0.04 (0.01)	Error Correction Term ⁺		-0.60 (0.05)
Industry-wide costs		0.97 (0.05)	Error Correction Term ⁻		-0.88 (0.05)
Industry-wide costs x PD		-0.09 (0.03)	Error Correction T. ⁺ x PD		0.27 (0.03)
Firm-specific costs		0.84 (0.02)	Error Correction T. ⁻ x PD		0.02 (0.01)
Firm-specific costs x PD		0.09 (0.02)			
Constant		0.31 (0.01)			
R^2 within		0.84	R^2		0.41
R^2 between		0.90			
R^2 overall		0.84			
Observations		101,104	Observations		98,264

Note: Robust standard errors in parentheses.
Source: Based on Bittmann et al. (2020).

Third, increasing market shares of private labels and the development of alternative private label brand formats demand further theoretical and empirical research on the effects of private labels for retail competition and the performance of retail markets. Steiner (2004) discusses the issue in his paper on the nature and benefits of national brand and private label competition.

Fourth, not all products are regularly promoted. Food stores use a selection of goods that is repeatedly promoted. So-called signpost items are products of high consumer price sensitivity. Lal and Matutes (1994) discuss this without describing the selection of products. Hosken and Reiffen (2004) present a model to derive some properties of promoted products and SAP and GfK (2010) publish lists of signpost items and their price sensitivities.

Fifth, the Covid-19 pandemic has shifted food demand towards online retailing. This will alter food marketing to use more individual data, individually tailored promotions, and social media-based advertising. Developing and investigating optimisation methods based on artificial intelligence is an area of emerging research. Chintagunta et al. (2016) look at the potential of big data in marketing science. Culotta and Cutler (2016) propose a novel and fully-automated method for inferring attribute-specific brand perception ratings based on mining a brand's social connections on Twitter.

11.7. Keywords and Terms

- Brand
- Brand loyalty
- Bertrand
- Buying-in price
- Cartel
- Cherry picking
- Concentration ratio (CR)
- Cournot-Nash equilibrium
- European article number (EAN)
- Everyday low-price (EDLP)
- Fast-moving consumer goods (FMCG)
- Food away from home (FAFH)
- High-Low or HiLo strategy
- Lerner index
- Loss-leader

- Market power
- Mark-up
- Menu costs
- Mixed bundling
- Mixed strategy
- Monopolistic competition
- Monopoly rent
- National brand (manufacturer brand)
- Odd pricing
- Price discrimination
- Price dispersion
- Price rigidity
- Product differentiation
- Promotional pricing (sales)
- Private label (retailer brand)
- Pure bundling
- Reaction function
- Residual demand (curve)
- Stackelberg
- Store format
- Store keeping unit (SKU)
- 9-endings

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11.9. Appendix: Derivation of Varian's (1980) model result.

$$F(P) = 1 - \left(\frac{k}{l} \left(\frac{1}{P} - \frac{1}{r} \right) \right)^{\frac{1}{n-1}} \quad (1)$$

$$\frac{\partial F(P)}{\partial P} = -\frac{1}{n-1} \left(\frac{k}{l} \left(\frac{1}{P} - \frac{1}{r} \right) \right)^{\frac{1}{n-1}-1} \left(-\frac{k}{lP^2} \right) \quad (2)$$

$$\frac{\partial F(P)}{\partial P} = \frac{1}{n-1} \left(\frac{k}{l} \right)^{\frac{1}{n-1}} \left(\frac{1}{P} - \frac{1}{r} \right)^{\frac{1}{n-1}-1} \frac{1}{P^2} \quad (3)$$

Let $m = 1 - 1/(n - 1)$. It holds that $n = rM/k$. Thus $m = (rM - 2k)/(rM - k)$

$$\frac{\partial F(P)}{\partial P} = \frac{k}{rM-k} \left(\frac{k}{l} \right)^{m-1} \left(\frac{1}{P} - \frac{1}{r} \right)^{-m} \frac{1}{P^2} \quad (4)$$

With $n \rightarrow \infty$, $m \rightarrow 1$ and $k/(rM - k)$ is a constant factor it follows:

$$\frac{\partial F(P)}{\partial P} \xrightarrow{n \rightarrow \infty} \left(\frac{1}{P} - \frac{1}{r} \right)^{-m} \frac{1}{P^2} \quad (5)$$

$$\frac{\partial F(P)}{\partial P} \stackrel{n \rightarrow \infty}{\approx} \frac{1}{\left(\frac{r-p}{rP}\right)^m p^2} \quad (6)$$

$$\frac{\partial F(P)}{\partial P} \stackrel{n \rightarrow \infty}{\approx} \frac{1}{\left(\frac{r-p}{r}\right)^m p^{2-m}} \quad (7)$$

$$\frac{\partial F(P)}{\partial P} \stackrel{n \rightarrow \infty}{\approx} \frac{1}{\left(1-\frac{p}{r}\right)^m p^{2-m}} \quad (8)$$

$$\frac{\partial F(P)}{\partial P} \stackrel{n \rightarrow \infty}{\approx} \frac{1}{P\left(1-\frac{p}{r}\right)} \quad (9)$$

11.10. Exercises

1. Bartels and Langness (BeLa) is a regionally operating food retailer in Northern Germany. Write up a small portrait of the company (<https://www.bela.de/en/>).
2. How have consumer trends and attitudes changed during the Covid-19 pandemic? Which of these changes are likely short-term, and which are likely long-term?
3. List some examples of monopolistic price setting in (German) food retailing and discuss graphically the welfare effects of a monopoly that is more cost efficient than the supply under perfect competition.
4. What concepts are used to measure the degree of market power in food retailing?
5. List some examples of monopolistic competitive price setting in food retailing.
6. What approaches are used to estimate the marginal costs of retailing?
7. Discuss the welfare effects of monopolistic competitive price setting in food retailing.
8. Summarise the main challenges of multiproduct pricing.
9. Visit the webpage of the German Antitrust Commission and summarise one of the cases on illegal cartel actions in the field of food retailing or food processing (<https://www.bundeskartellamt.de/EN/>).

10. Find examples for food retail product price discrimination and estimate relative mark-up differences.
11. Draw a graph of an individual demand function and explain the implementation of a take-it-or-leave-it price or a two-tier pricing scheme, and the effects on welfare of first-degree price discrimination.
12. Describe several real-world examples of two-tier pricing schemes.
13. For the pizza example under 'second-degree price discrimination' in Section 4.6: Maximise the profit and calculate the optimal size for the small pizza, when $MWP_1=100-2q$ und $MWP_2=100-q$. Discuss how the results change if the shares of pizza lovers change from 1:1 to 2:1 or to 1:2.
14. Explain why the aggregated marginal revenue of two or more markets is not necessarily similar to the marginal revenue of the aggregated demand over the same markets.
15. Discuss potential differences in the market outcomes between uniform monopoly pricing and third-degree price discrimination.
16. Discuss examples of bundling in food retailing, classify the type of bundling in each case, and explain the motivation for the bundling.
17. Test your knowledge of food retail prices. Write down 10 items (e. g. 1kg bananas, 1L milk, etc.) and their prices. Check your data against prices at an online retailer such as Rewe.
18. Read the paper by Varian (1980) and summarise his model assumptions. Which assumptions differ from Salop and Stiglitz (1977)? What are the main findings and shortcomings of Varian's (1980) model?
19. Set the expected profit in equation (52) equal to zero and derive $F(P)$ in equation (53).
20. In Varian's (1980) model, stores discriminate between informed and uninformed consumers. Explain this statement and describe the type of discrimination.
21. Read the paper by Chevalier et al. (1994) and describe the variables used in their empirical analysis. What are the main results of their analysis, and how do they support the overall conclusion of the paper?

22. Compare the results presented in Figure 11.25 and your personal observations with the results found by Berck et al. (2008).